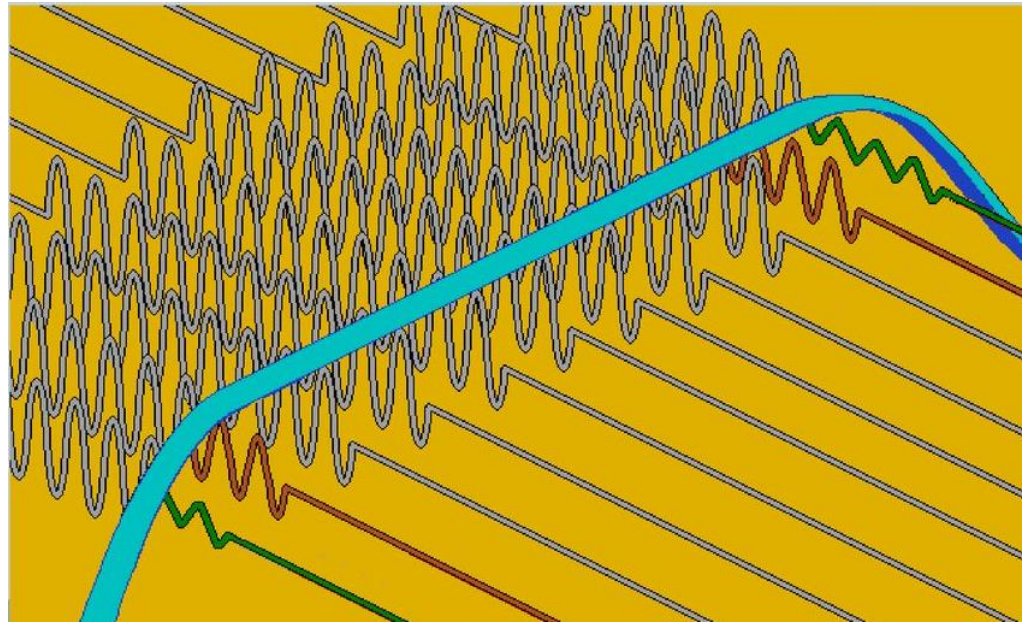


TTK4240 Industriell elektroteknikk

January 13, 2025



Marta Molinas

Institutt for teknisk kybernetikk

Lecture - Week 03 – Passive Filters Part I

- **Low Pass Filters**

- Series RL, RC
- Cutoff frequency
- Transfer Function

- **High-Pass Filters**

- Series RC
- Cutoff frequency
- Transfer Function

- **Bandpass Filters**

- Center Frequency, Bandwidth and Quality Factor
- Series RLC
- Relating Frequency domain and time domain

- **Bandreject Filters**

- Series RLC
- Transfer Function

Nilsson's Book Chapter 14: 14.1 to 14.5

Overview of *s*-domain Analysis

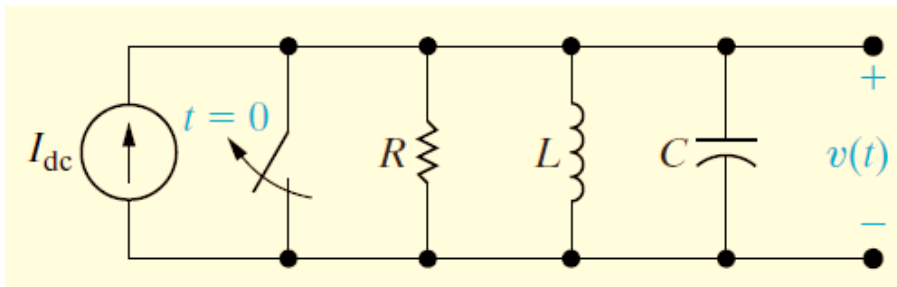
The Transfer-function

Definition

- The transfer-function $H(s)$ is a *s-domain* ratio of the output function $Y(s)$ to the input function $X(s)$, which can be generally formulated as

transfer-function

$$H(s) = \frac{Y(s)}{X(s)}$$



$$\frac{V(s)}{I_{dc}(s)} = H(s)$$

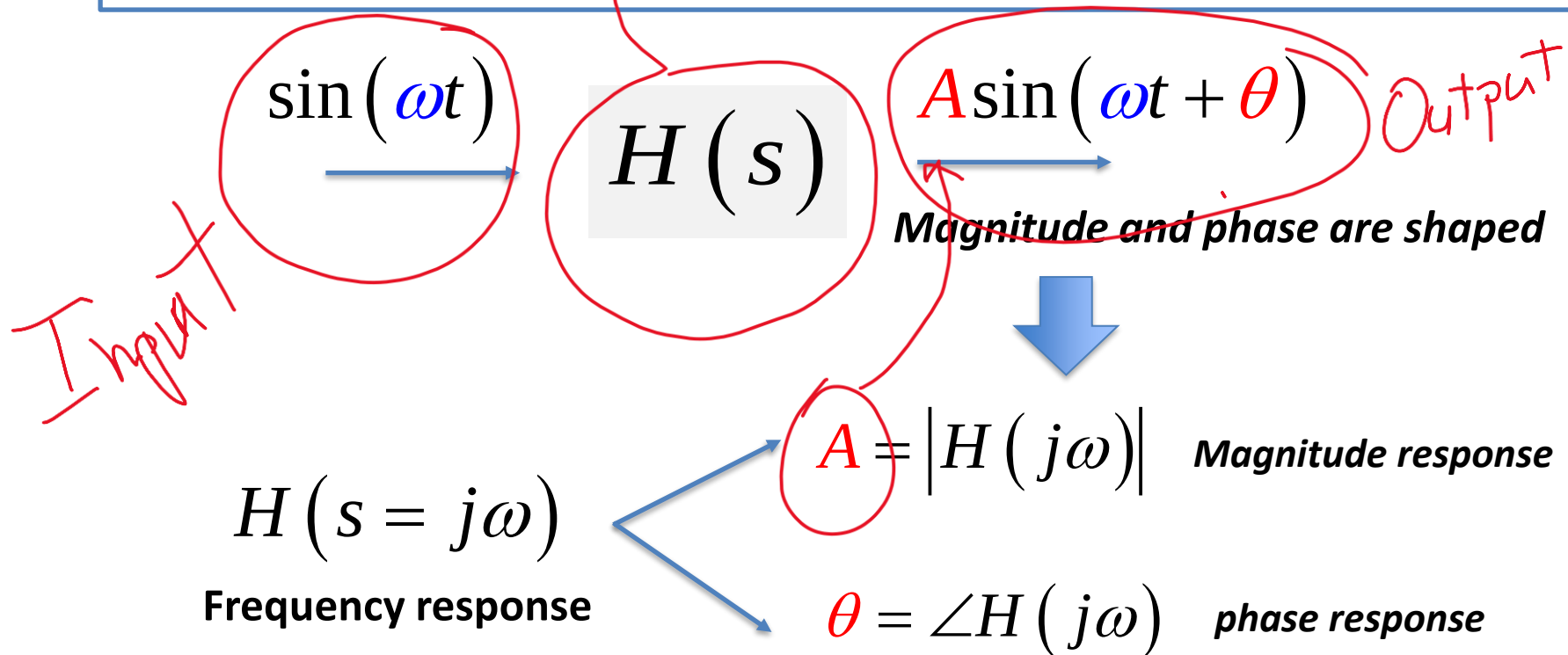
$$H(s) = \frac{1}{sC + 1/(sL) + 1/R}$$

Impedance

The Frequency response

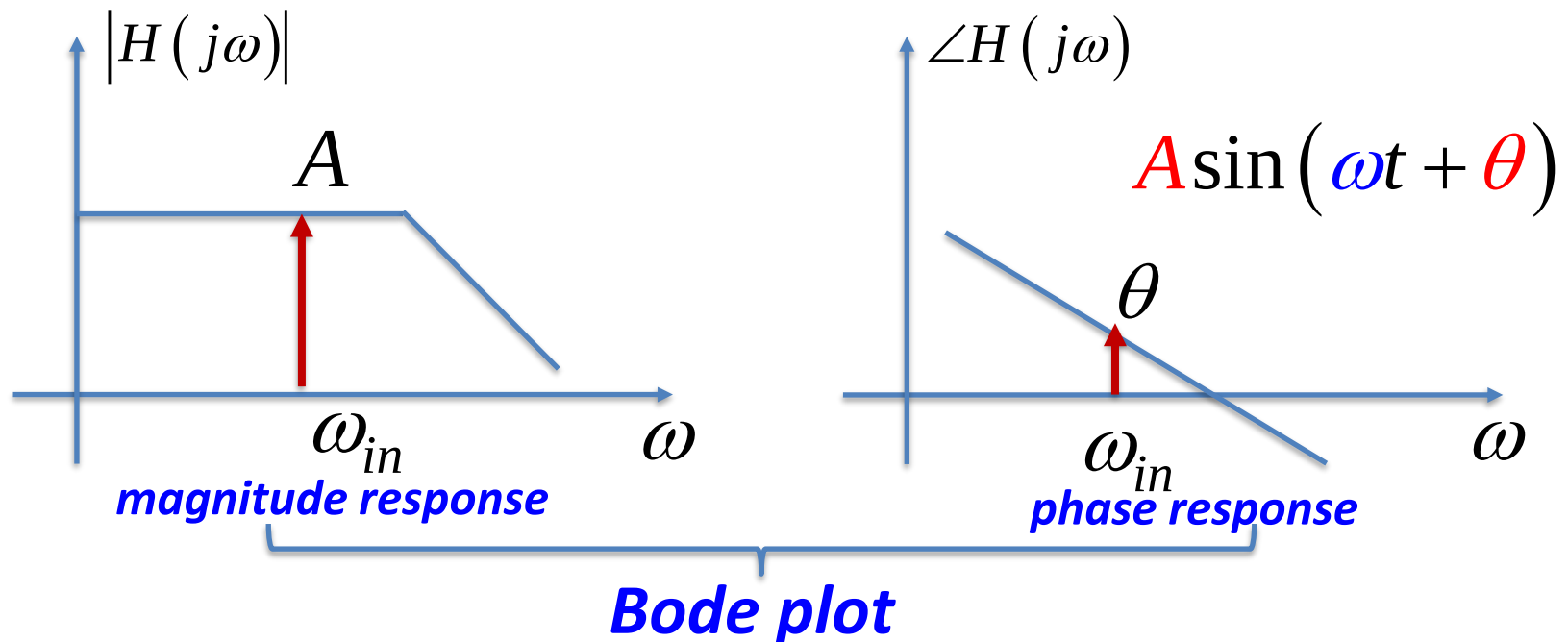
Review

- The frequency response of a transfer function shows how an input signal at a specified frequency will be shaped.



Bode plot

- The **Bode plot** is a graph usually used to analyze the frequency response of a system. It contains the plots of the *magnitude response* and *phase response*, which are all evaluated with varying frequency inputs.



Frequency-dependant circuit

- A **circuit is frequency-dependent** because most of the **passive circuit elements are frequency dependent** except for the resistor.
- Physically, if the circuit element can store energy (i.e. has memory), usually it is frequency-dependent.

	<i>Resistor</i>	<i>Inductor</i>	<i>Capacitor</i>	
<i>Time domain:</i>	R	L	C	
<i>s-domain:</i>	R	sL	$\frac{1}{sC}$	
 $s = j\omega$				
<i>Frequency-domain:</i>	R	$j\omega L$	$\frac{1}{j\omega C}$	<i>Frequency dependent</i>

Impedance or admittance

Definition

- The *impedance is* a *s-domain* ratio of voltage $V(s)$ to current $I(s)$, which can be generally formulated as

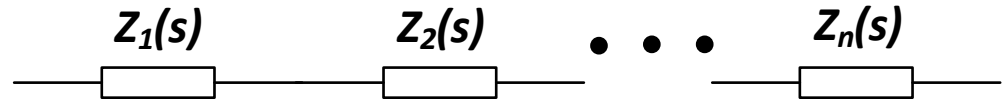
$$Z(s) = \frac{V(s)}{I(s)}$$

- Its inverse relationship, is referred to as the *admittance*

$$Y(s) = Z^{-1}(s) = \frac{I(s)}{V(s)}$$

Series and parallel connections of impedances

- **Series connection**

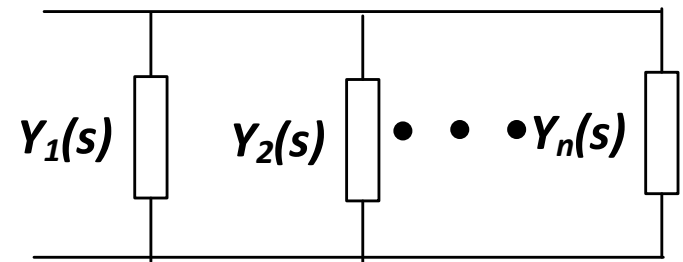


$$Z_{series}(s) = Z_1(s) + Z_2(s) + \dots + Z_n(s)$$

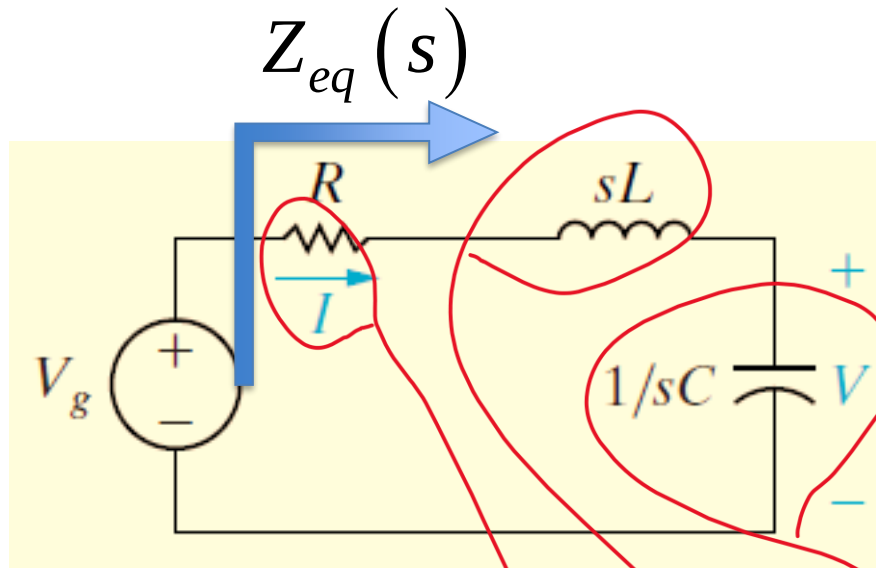
- **Parallel connection**

$$Y_{parallel}(s) = Y_1(s) + Y_2(s) + \dots + Y_n(s)$$

$$Z_{parallel}(s) = Y_{parallel}^{-1}(s)$$



A series connection example



Problem:
solve for $Z_{eq}(s)$

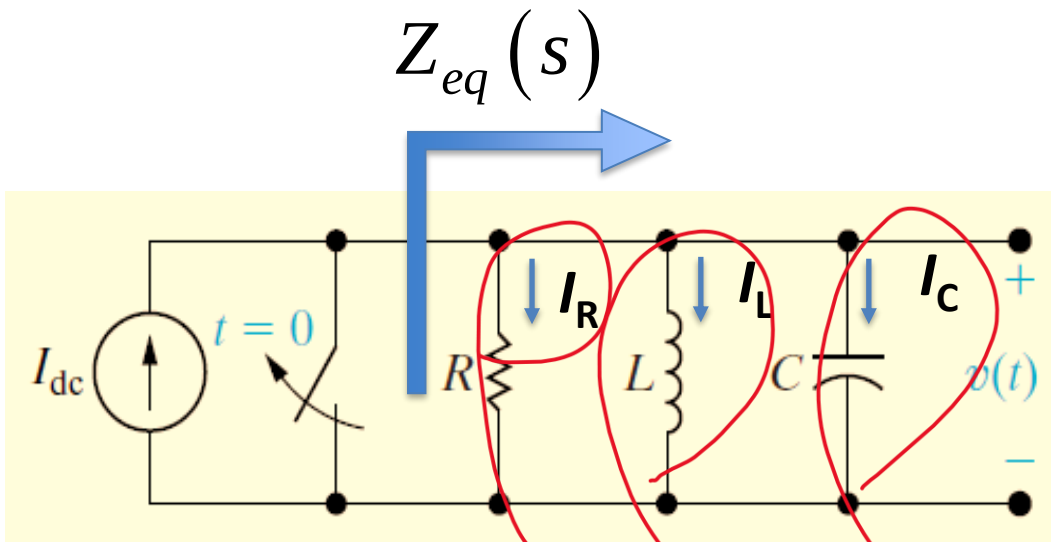
Method:
Apply the Ohm's law, or the Kirchhoff's **voltage** law

Equivalent impedance

$$\downarrow V_g = V_R + V_L + V_C$$

$$Z_{eq}(s) = \frac{V_g(s)}{I(s)} = \frac{RI(s) + sLI(s) + \frac{1}{sC}I(s)}{I(s)} = R + sL + \frac{1}{sC}$$

A parallel connection example



Problem:
solve for $Z_{eq}(s)$

Method:
Apply the Ohm's law, or the Kirchhoff's **current** law

$$I_{dc}(s) = I_R + I_L + I_C$$

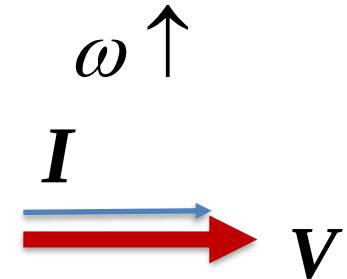
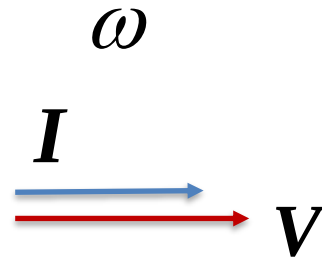
Equivalent impedance

$$Y_{eq}(s) = \frac{I_{dc}(s)}{V(s)} \quad Z_{eq} = Y_{eq}^{-1}(s) = \frac{1}{\frac{1}{R} + \frac{1}{sL} + sC} = \frac{1}{\frac{1}{R} + \frac{1}{sL} + sC}$$

Frequency-domain characteristics of R, L, and C: *Mag & phase*

Elements

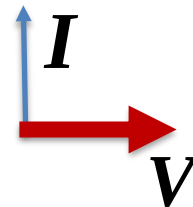
$$Z_R(s) = R$$



$$Z_L(j\omega) = j\omega L$$



$$Z_C(j\omega) = \frac{1}{j\omega C}$$



Frequency-domain characteristics of R, L, and C: *circuit perspective*

Elements

$\omega = 0$

$\omega = \infty$

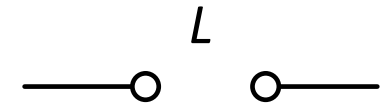
$$Z_R(s) = R$$



$$Z_L(j\omega) = j\omega L$$

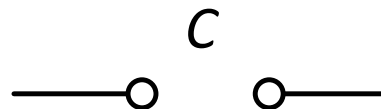


Short circuit

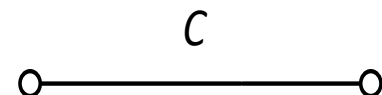


Open loop

$$Z_C(j\omega) = \frac{1}{j\omega C}$$

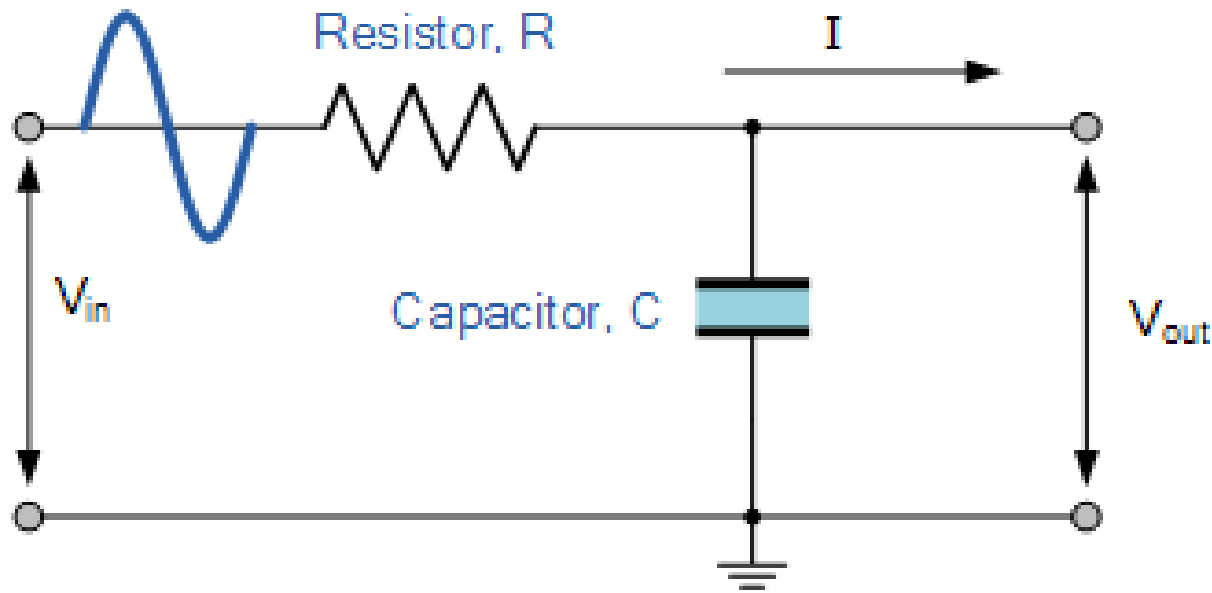


Open loop



Short circuit

What is a Filter?



$$\begin{array}{c} C \\ \text{---} \bigcirc \quad \bigcirc \text{---} \\ \omega = 0 \end{array}$$

$$V_{out} = V_{in}$$

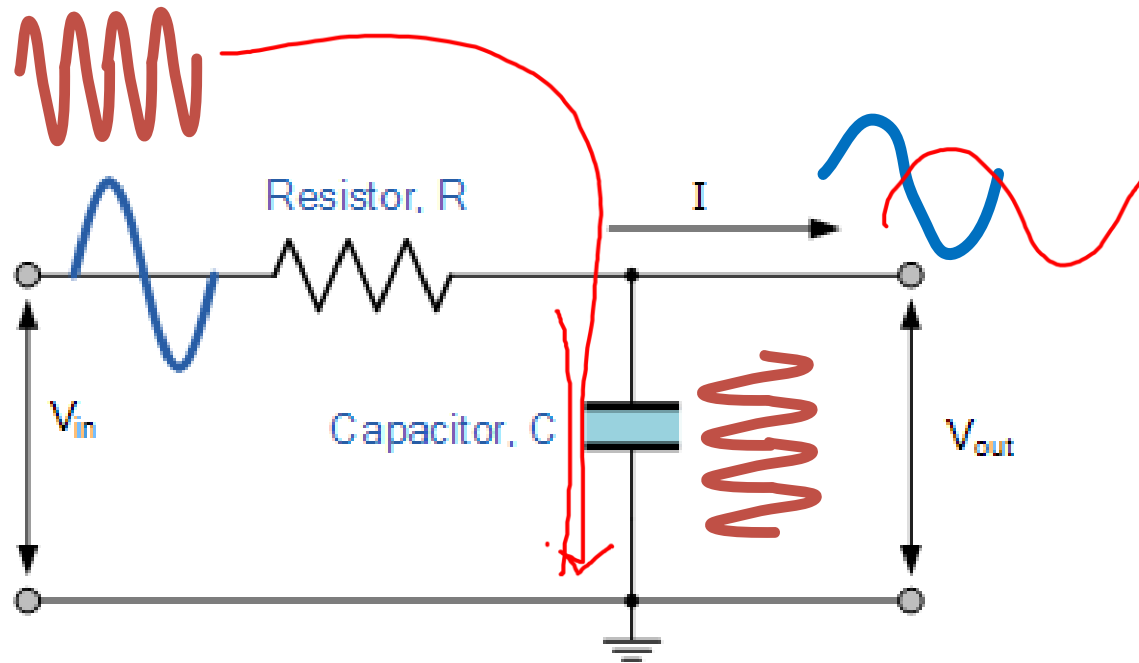
$$\begin{array}{c} C \\ \text{---} \bigcirc \quad \bigcirc \text{---} \\ \omega = \infty \end{array}$$

$$V_{out} = 0$$



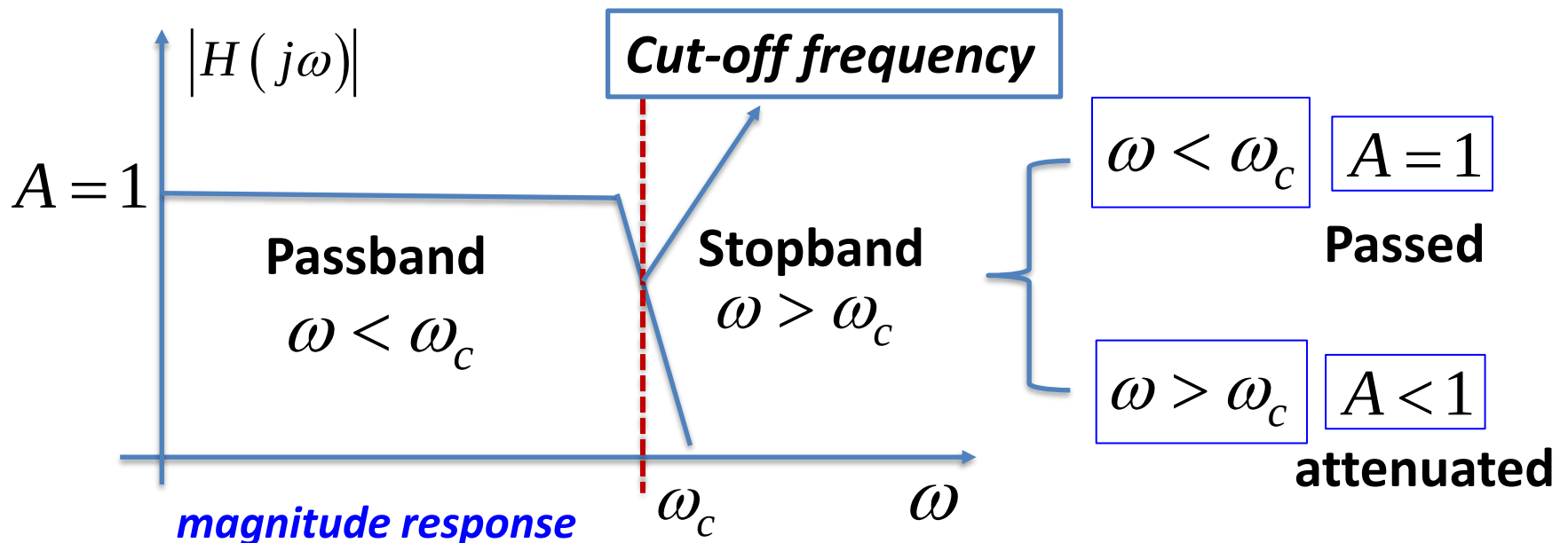
*Only allows inputs
at **low frequencies**
to reach the output*

Frequency selective behavior

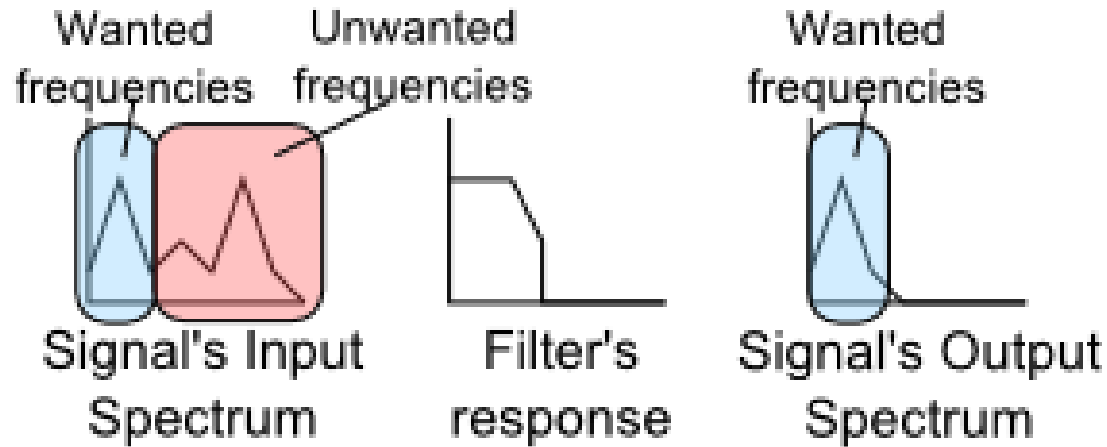


The Filter- Frequency Selective Circuits

The **Filter** (*strictly the passive filter*) is a **frequency-dependent circuit that can selectively pass** signals at certain frequencies, while attenuates signals at other frequencies



Where are Filters used?

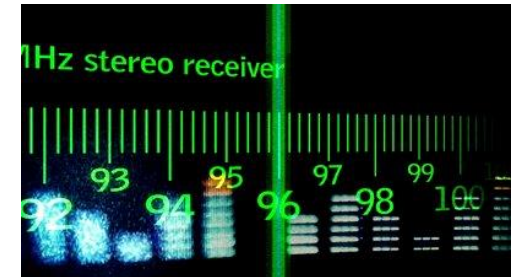
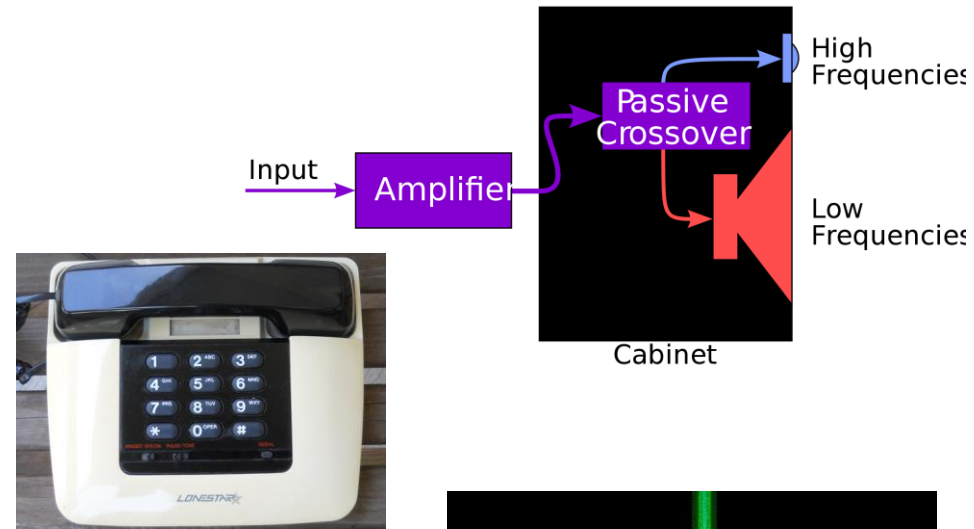


In radios, filters are used to select one radio station's signal while rejecting the signals from others transmitting at different frequencies.

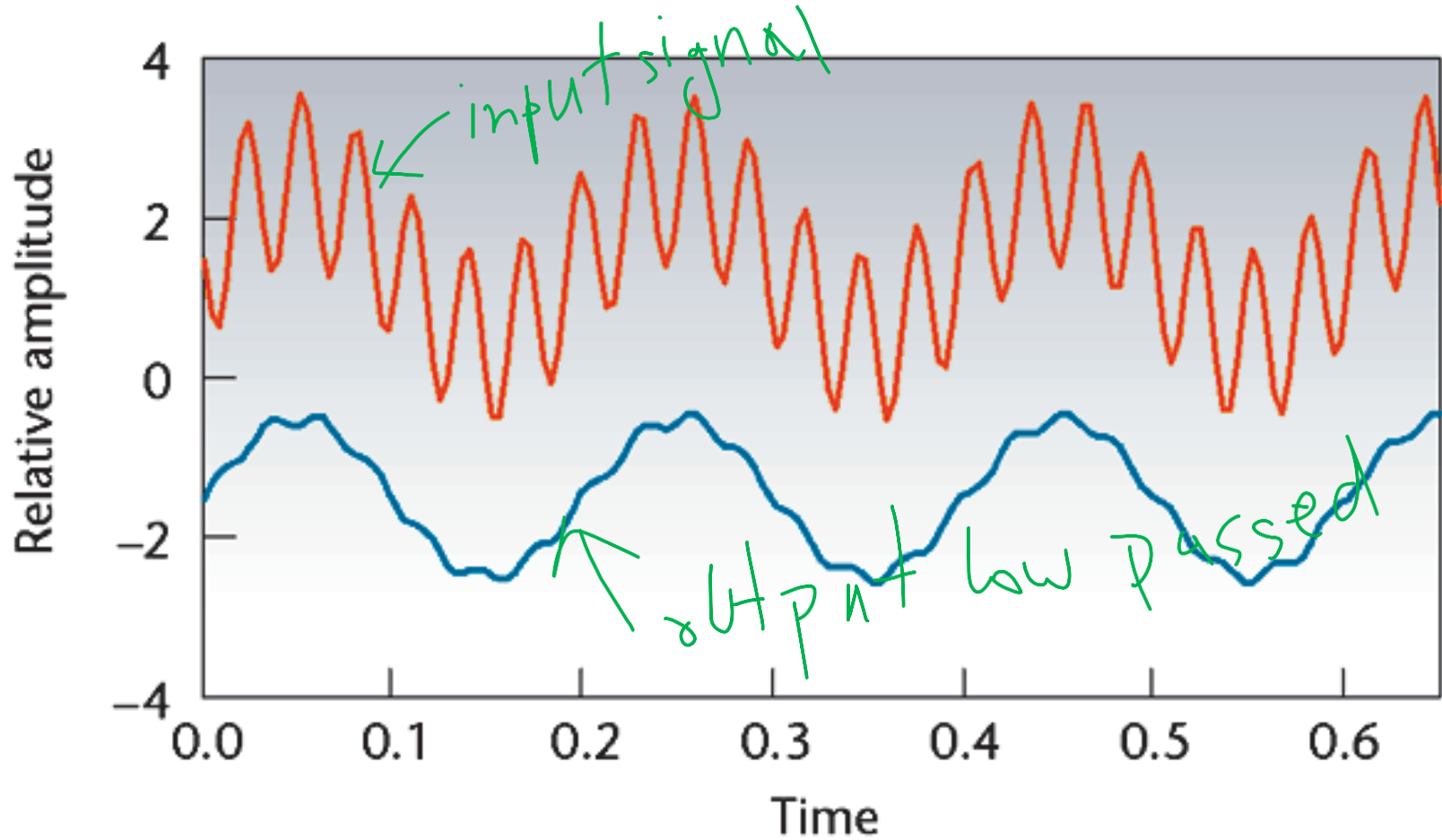
In stereo systems, filters are used to adjust the relative strengths of the low- and high-frequency components of the audio signal. Filters are also used throughout **telephone systems**.

Applications

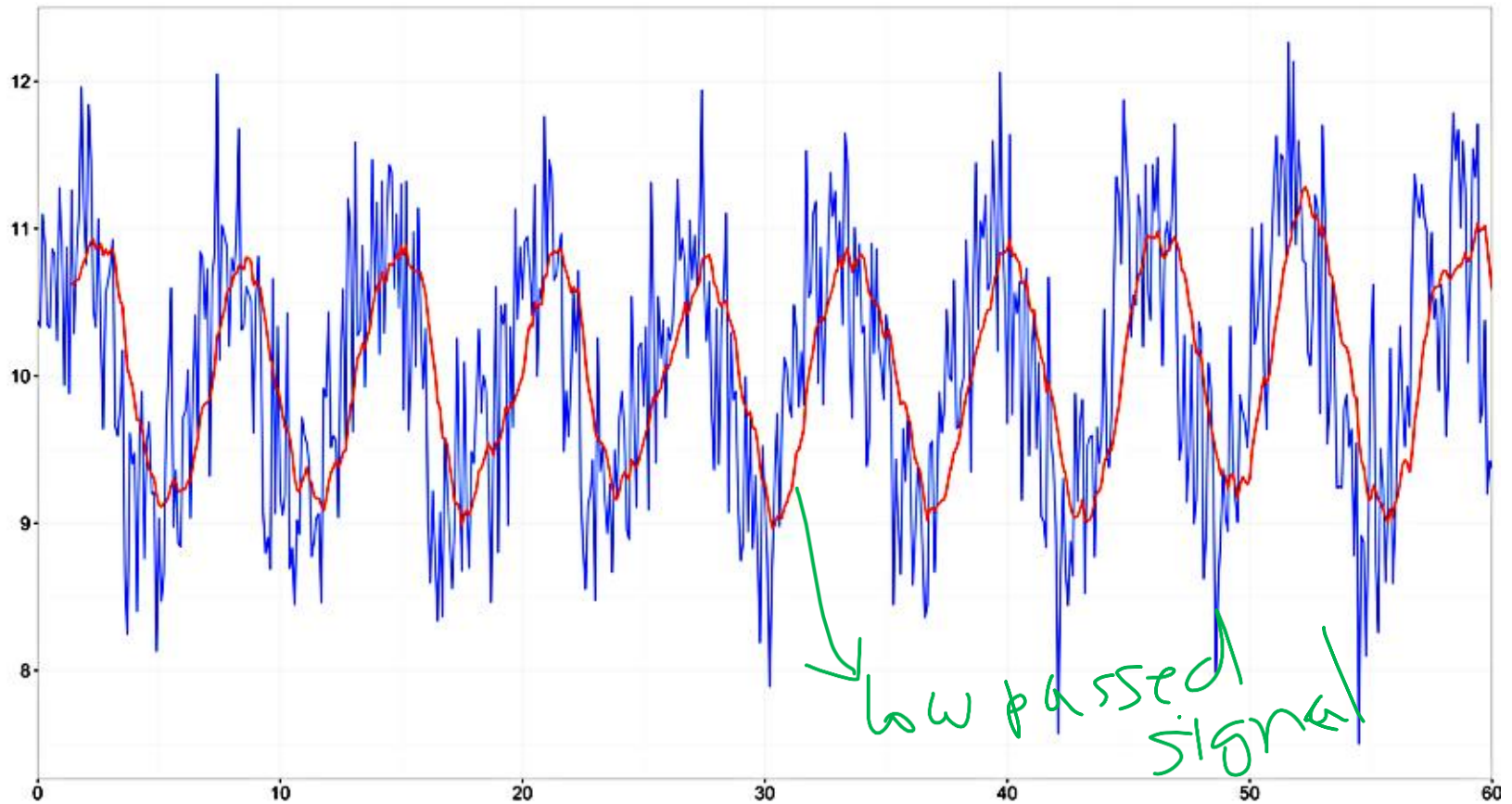
- Speaker crossover
- Power Filter
- Radio signals
- Audio signal, equalizer
- Biomedical signals (ECG, EEG, EOG)
- Sensors and signal processing
- Noise removal



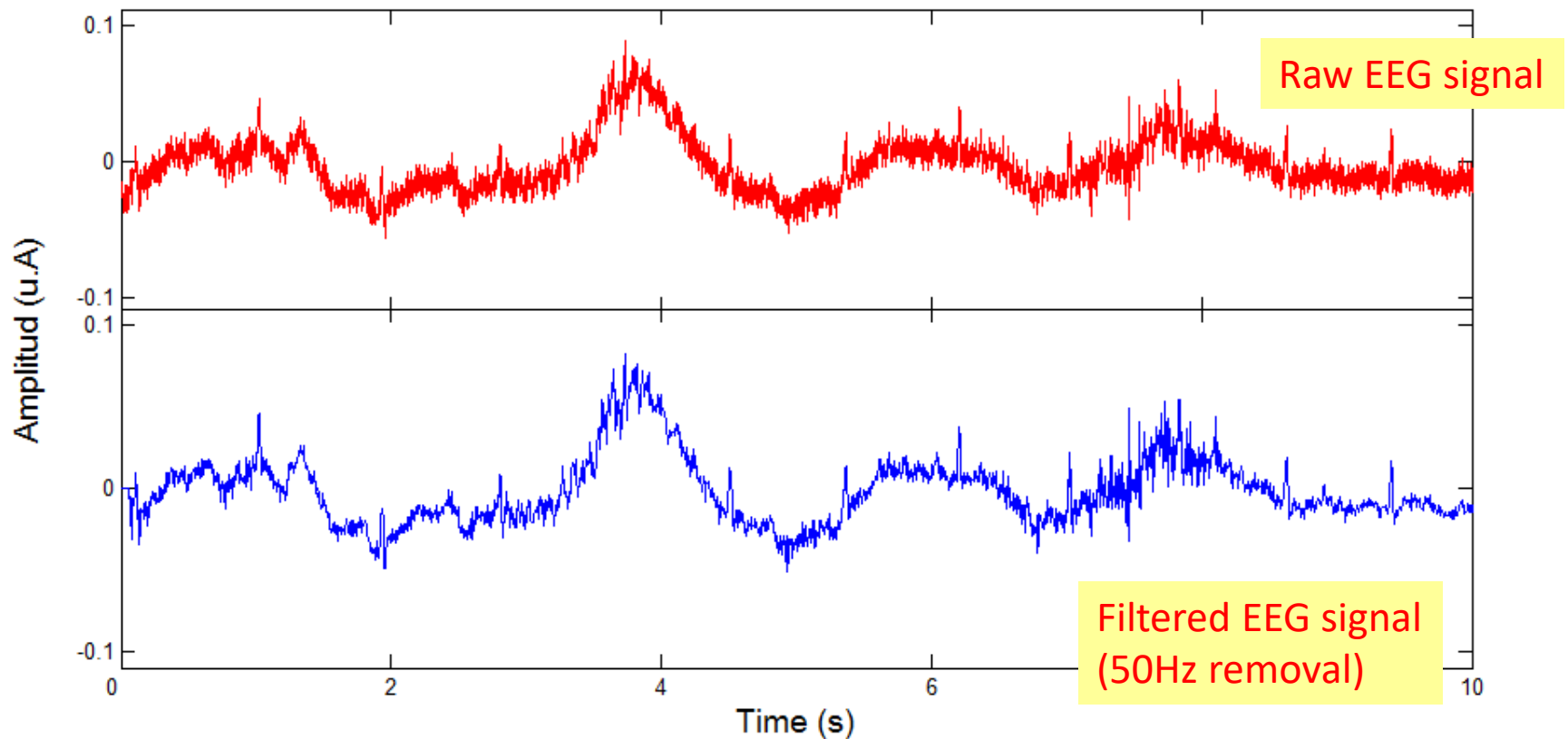
Example of Low Pass Filtering



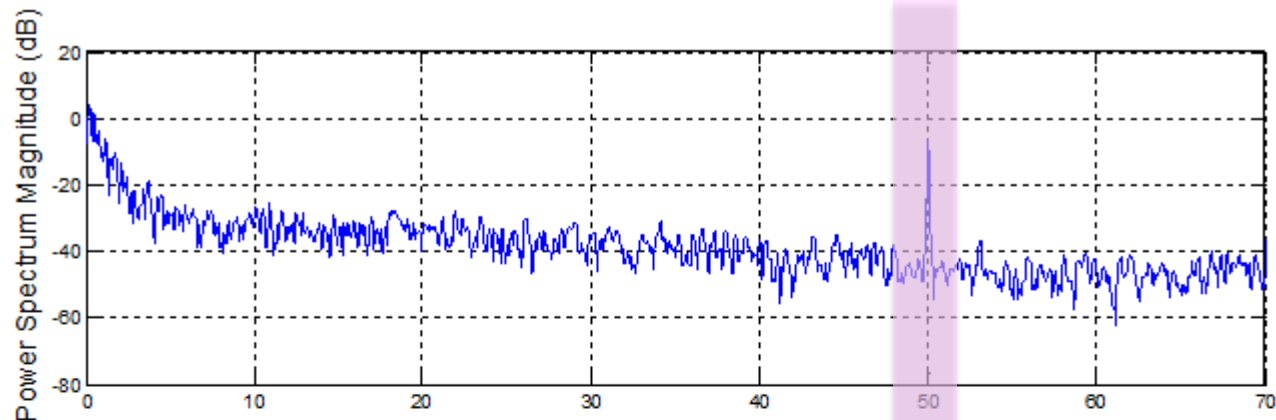
Filtering out the noise



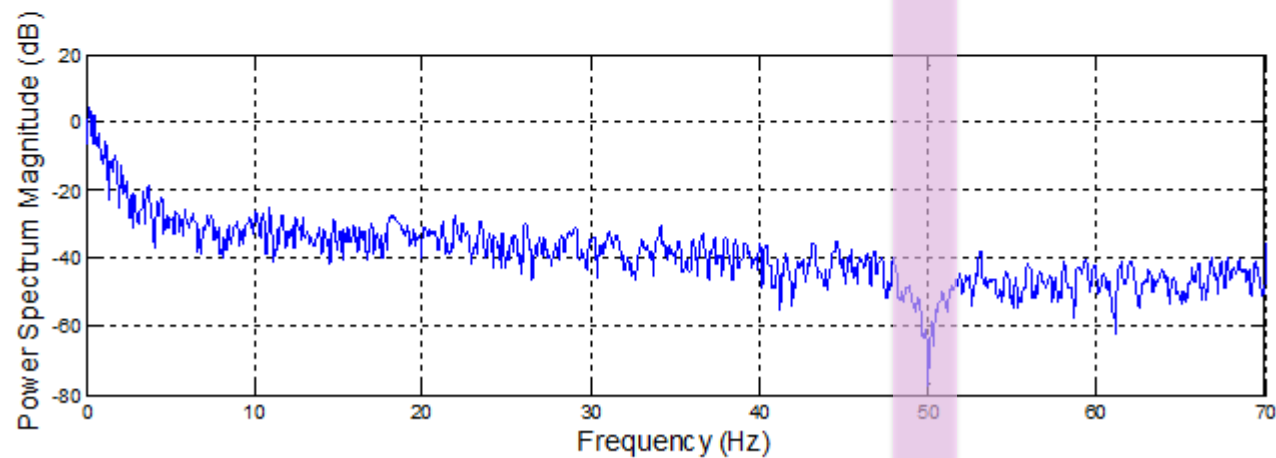
EEG Signals from the human brain



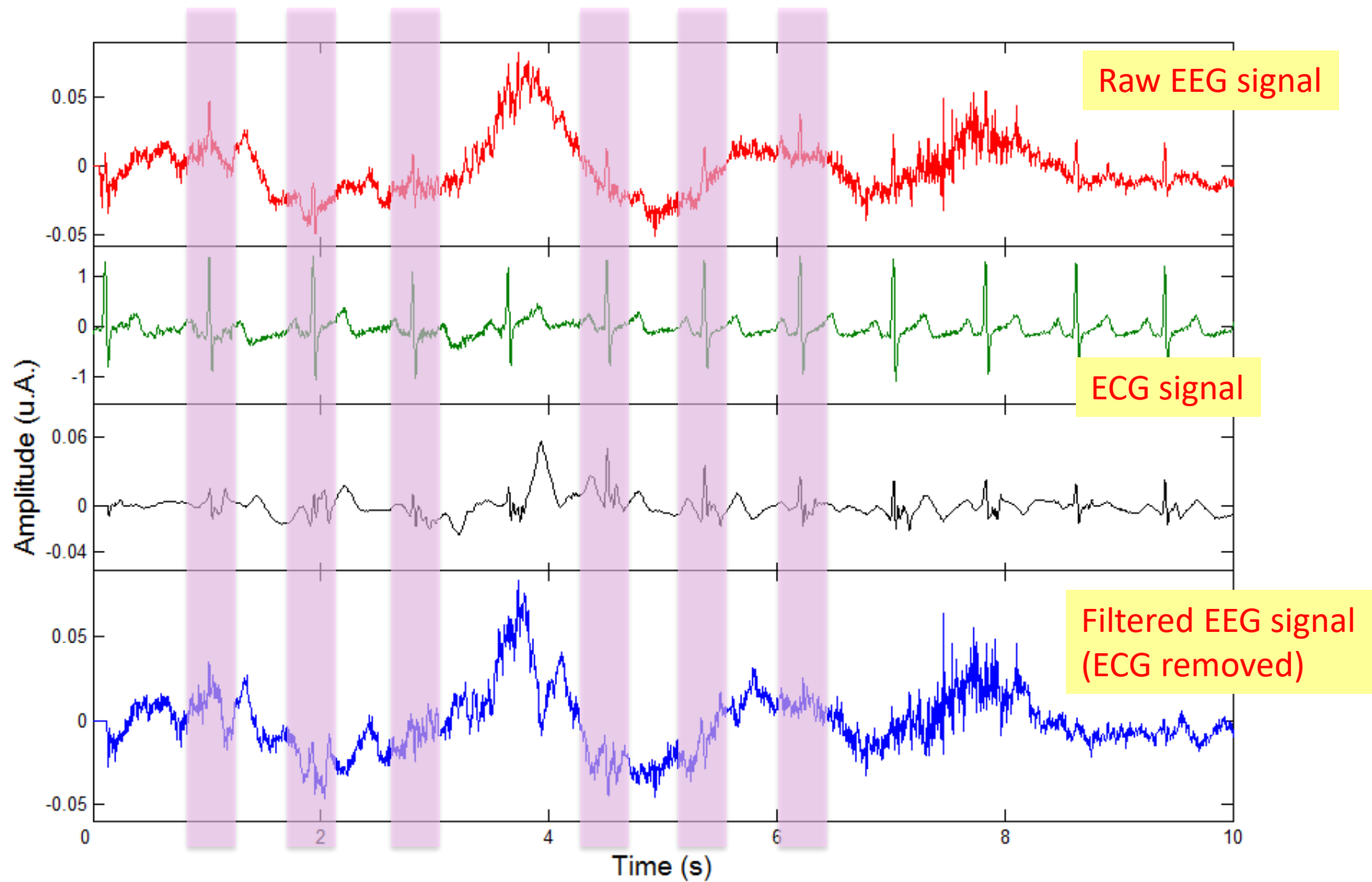
Power spectral density



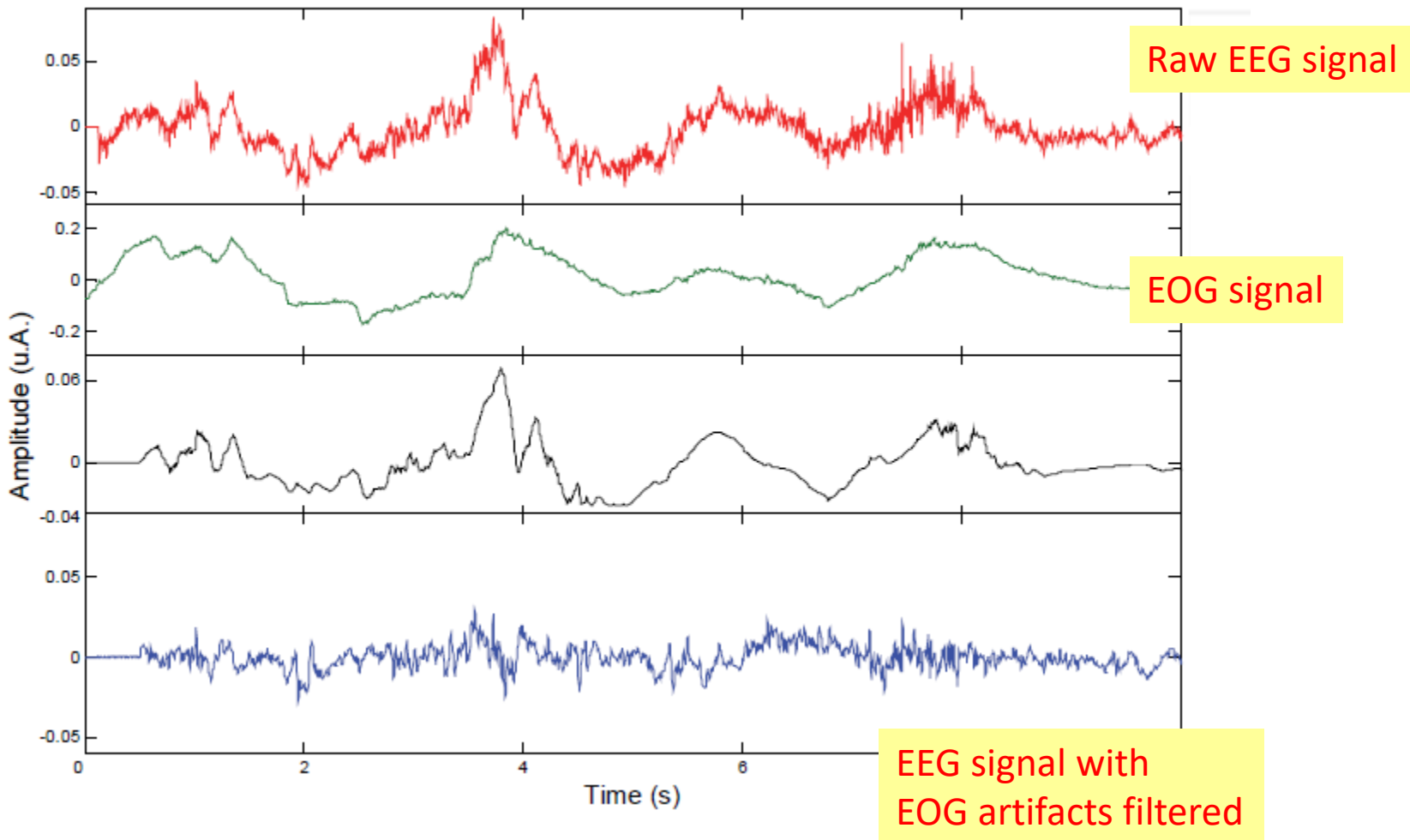
a)



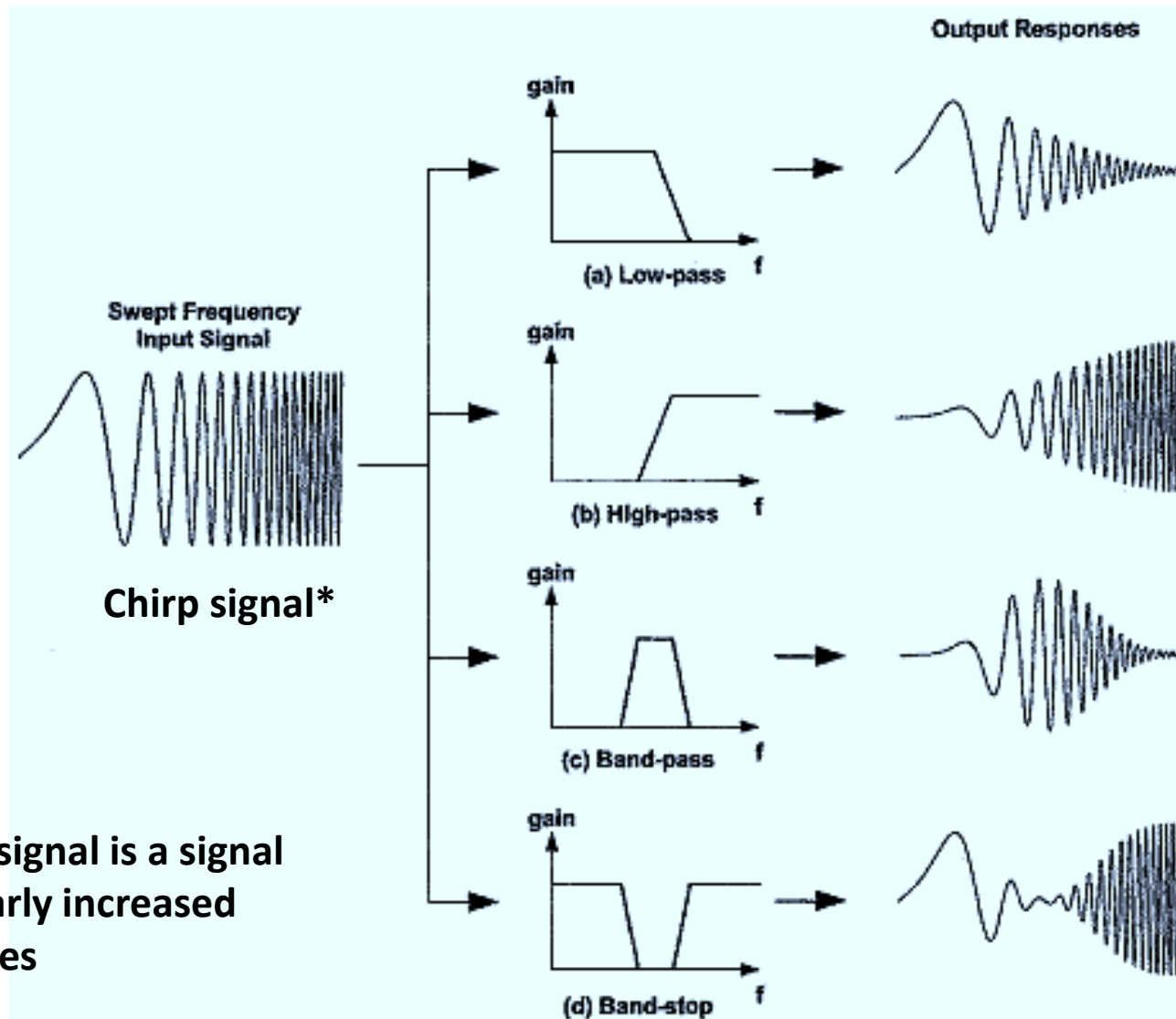
b)



EEG Signals from the human brain



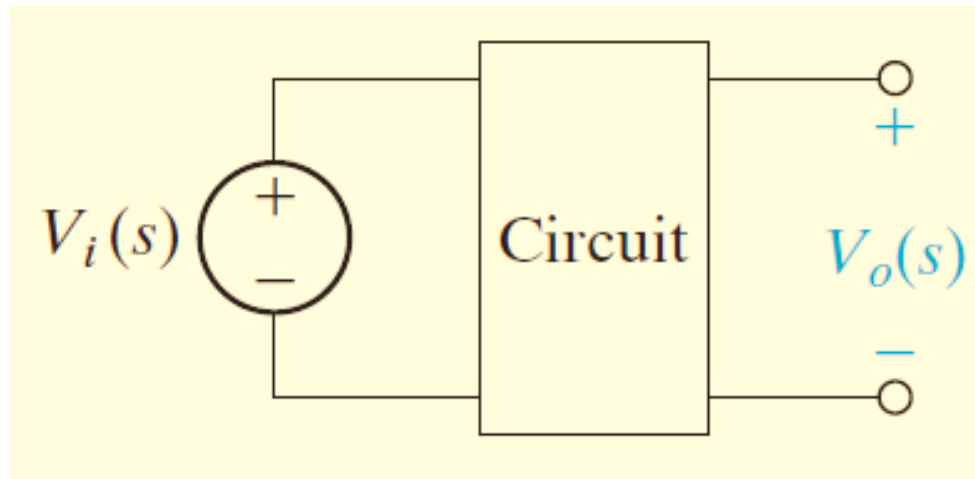
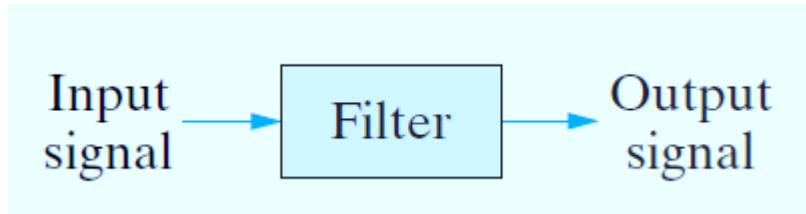
Various Filtering Actions



* A chirp signal is a signal with linearly increased frequencies

Break

The Action of a Filter

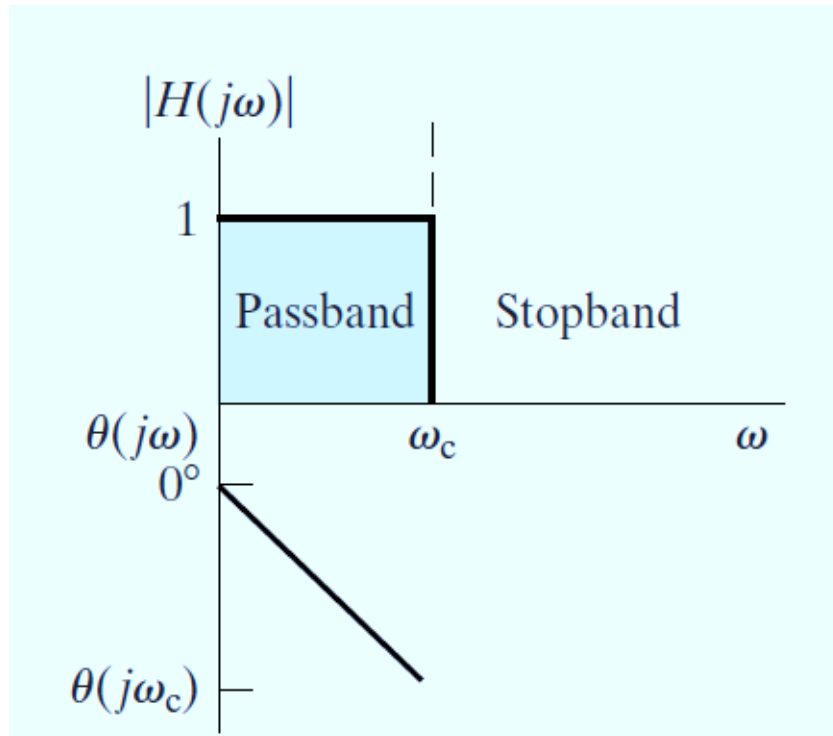


A circuit with voltage input and output

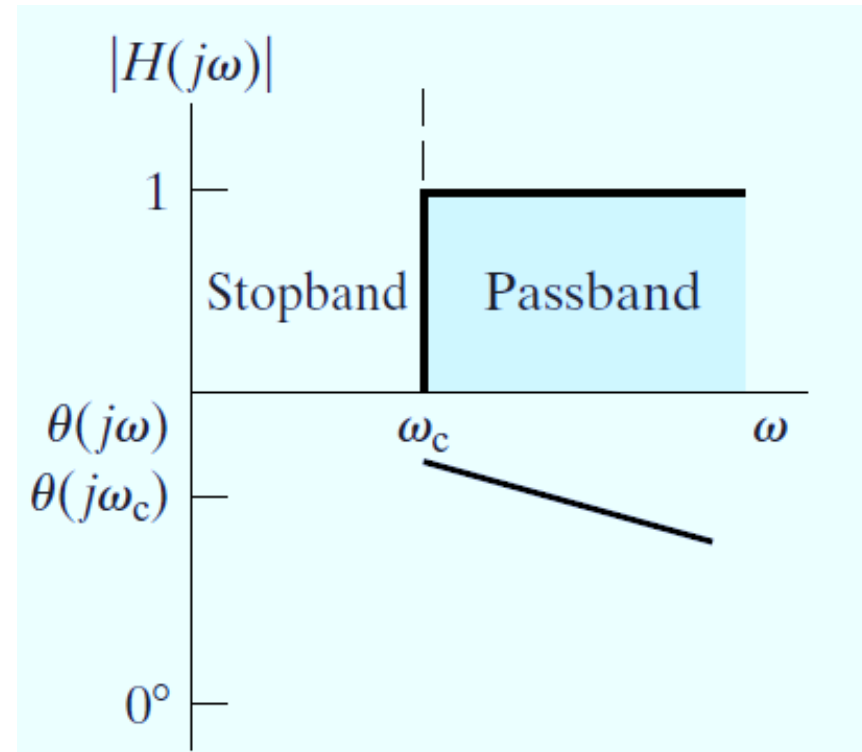
$$H(s) = V_o(s)/V_i(s).$$

The selectivity of the circuit (i.e. the filter) can be intuitively analyzed by its frequency response $H(j\omega)$

Ideal Frequency Responses

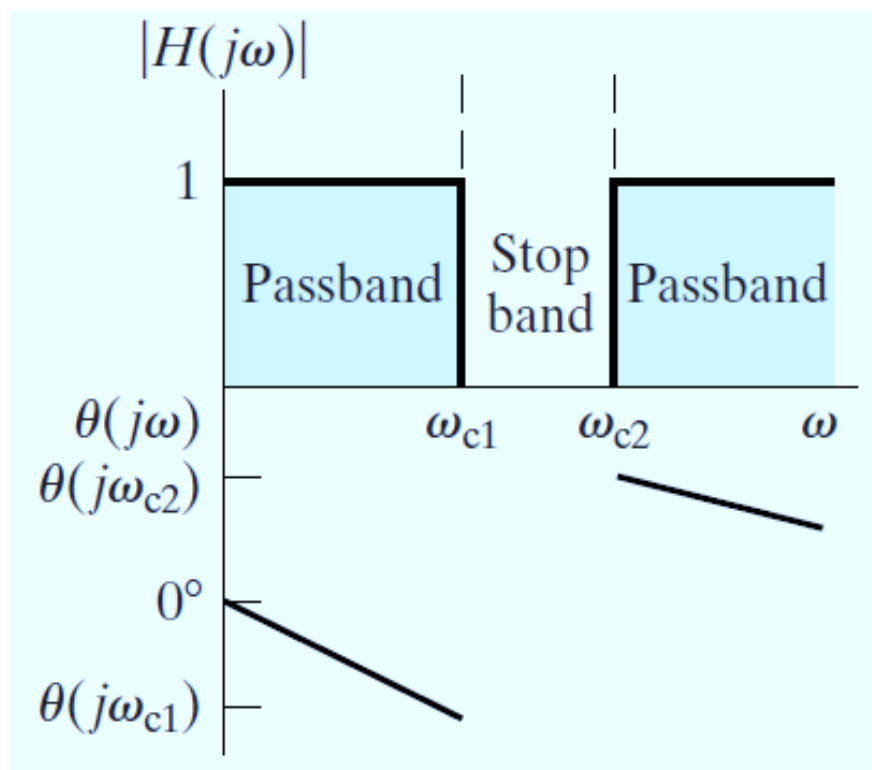


Ideal Low-pass filter

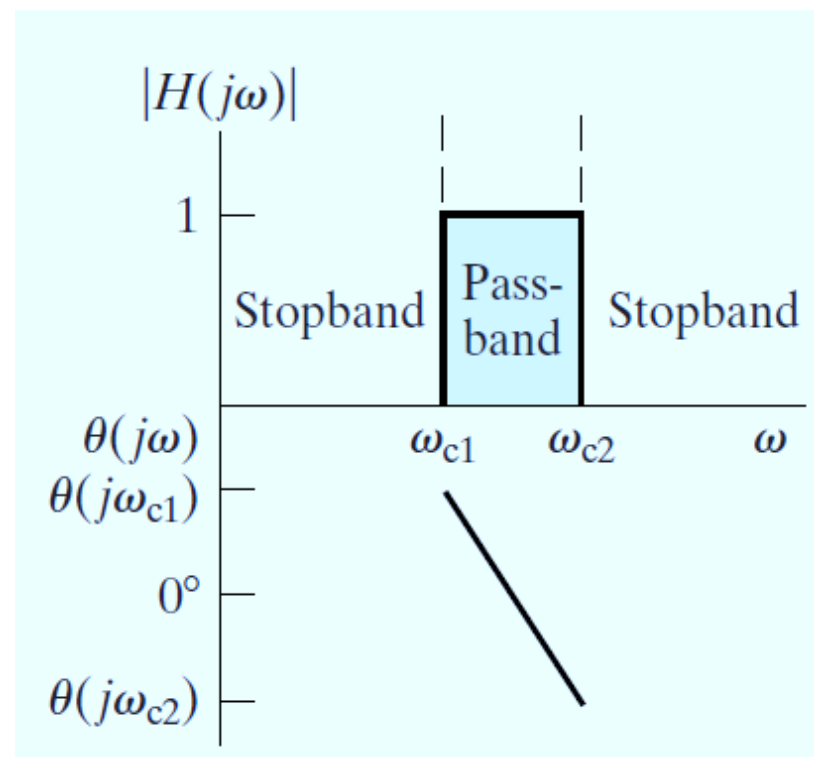


Ideal High-pass filter

Ideal Frequency Responses

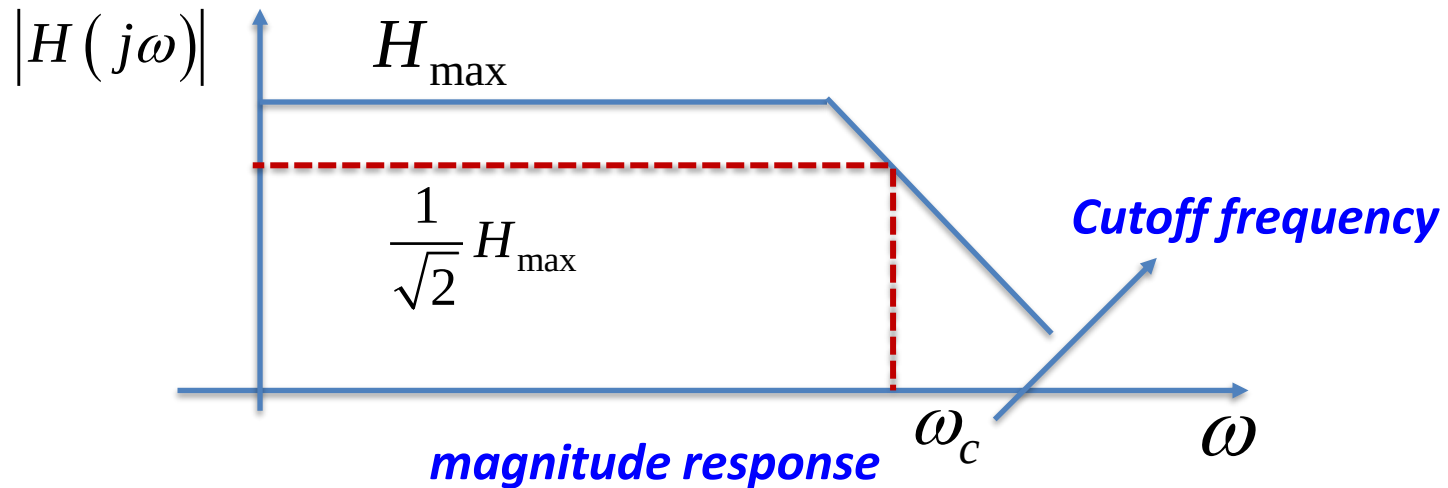


Ideal Bandreject filter



Ideal Bandpass filter

The Cutoff Frequency ω_c of practical passive filters



- The cutoff frequency is the frequency at which the magnitude of the transfer function equals:

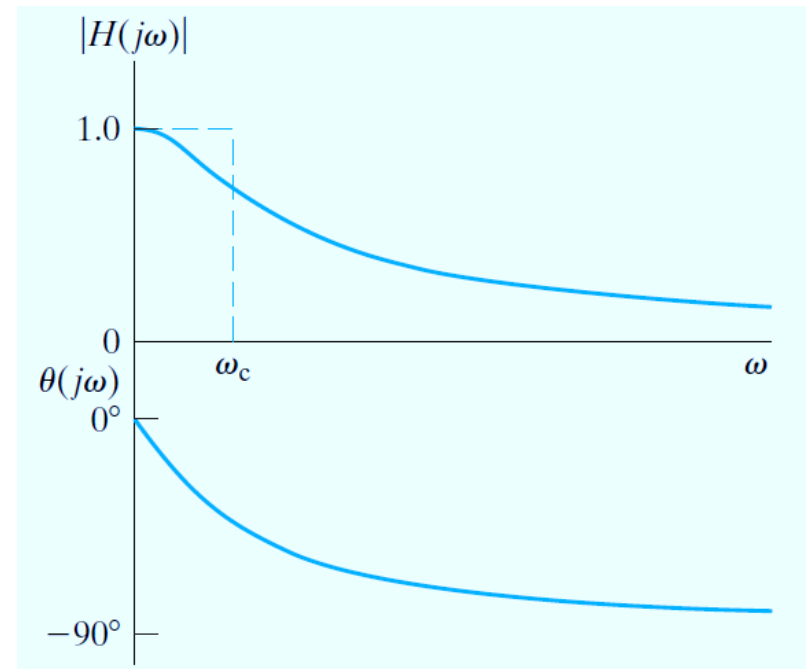
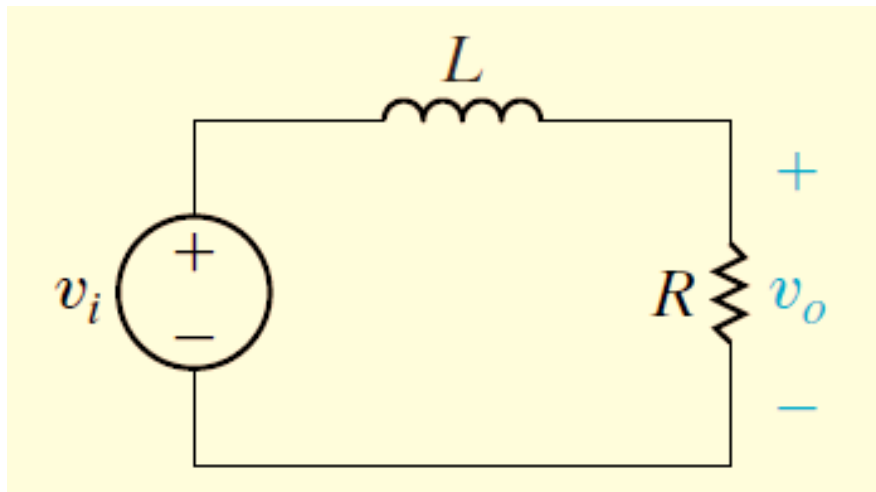
$$\left(1/\sqrt{2}\right)H_{\max}$$

Low-pass Filter

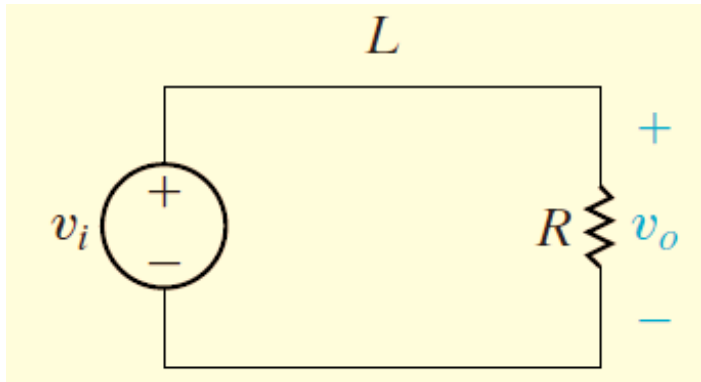
Hand notes

Low-Pass Filters

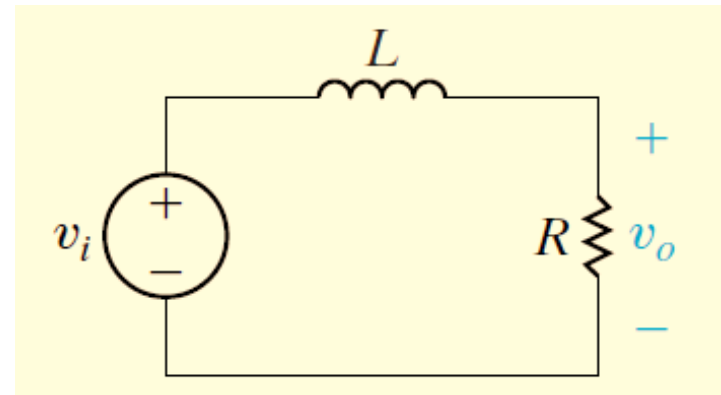
A RL circuit which will pass low frequencies and filter out or attenuate signals of high frequency (defined by the cutoff frequency)



Low-Pass Filters

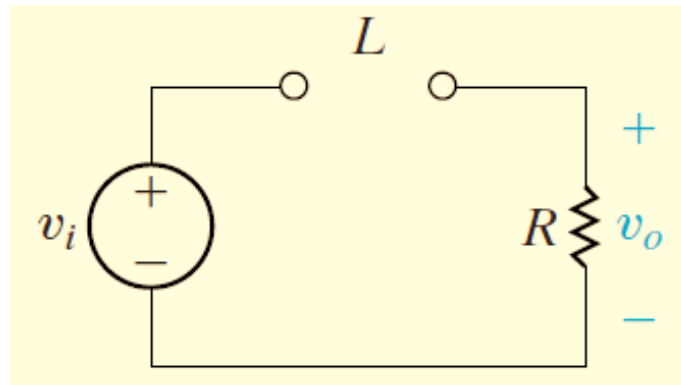


For $\omega=0$



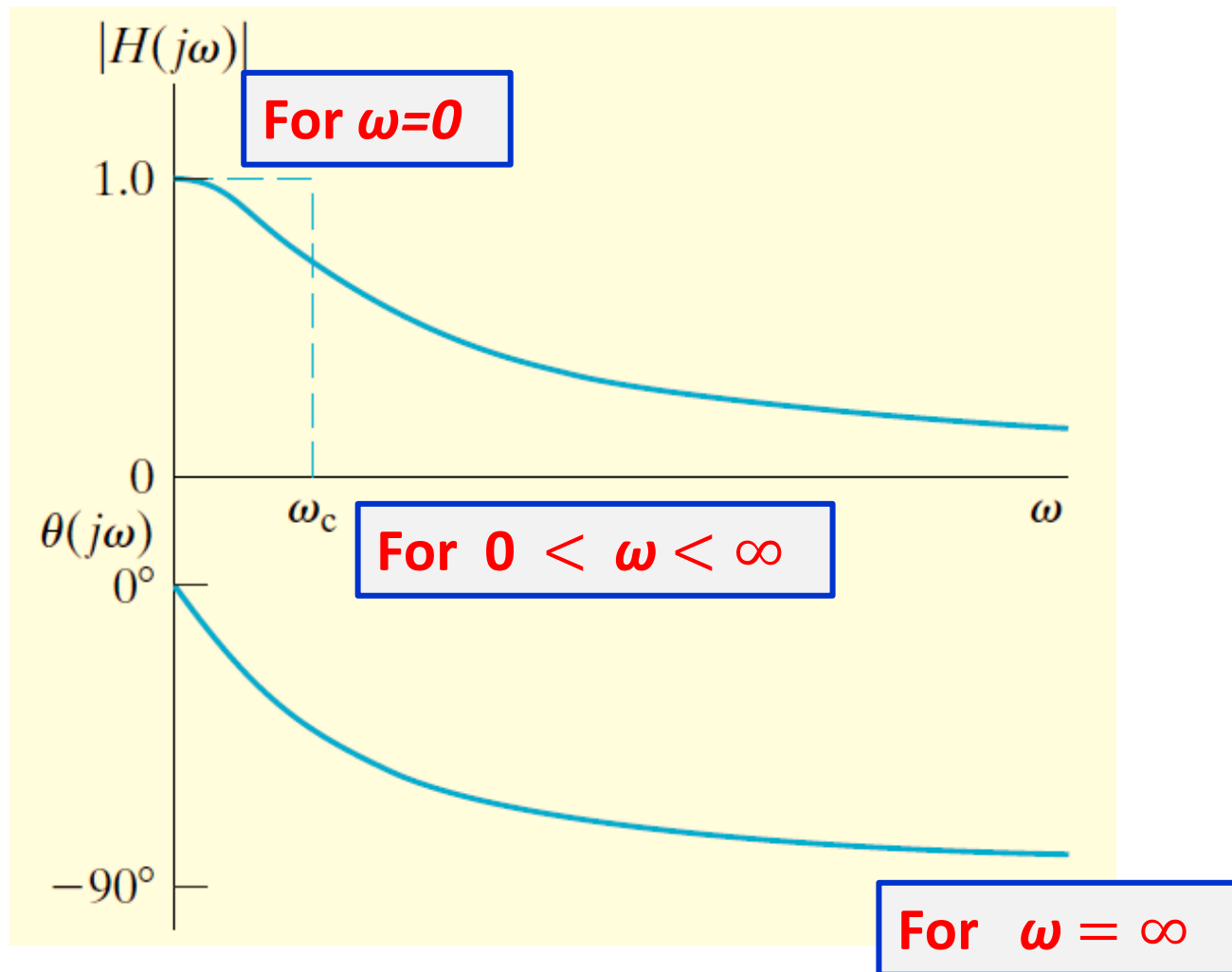
For $0 < \omega < \infty$

Low frequencies
refer to $\omega L \ll R$
High frequencies
refer to $\omega L \gg R$

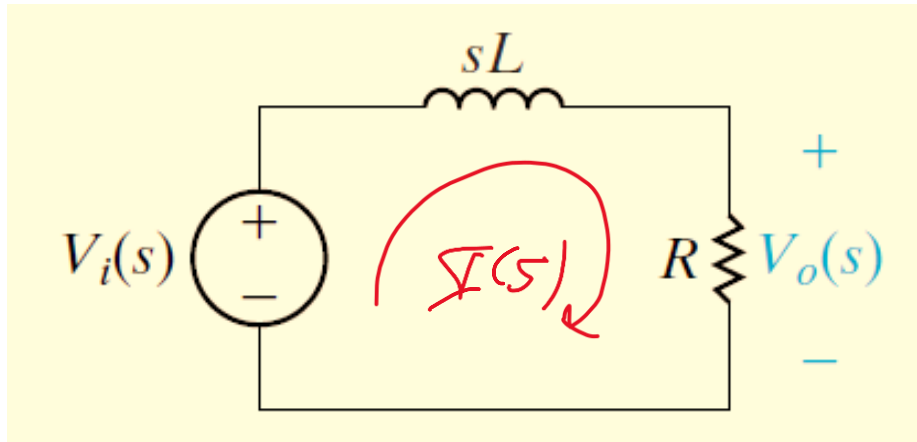


For $\omega = \infty$

Magnitude and Phase Angle Plots



Low-Pass Filter: Transfer Function



$$H(s) = \frac{R/L}{s + R/L}$$

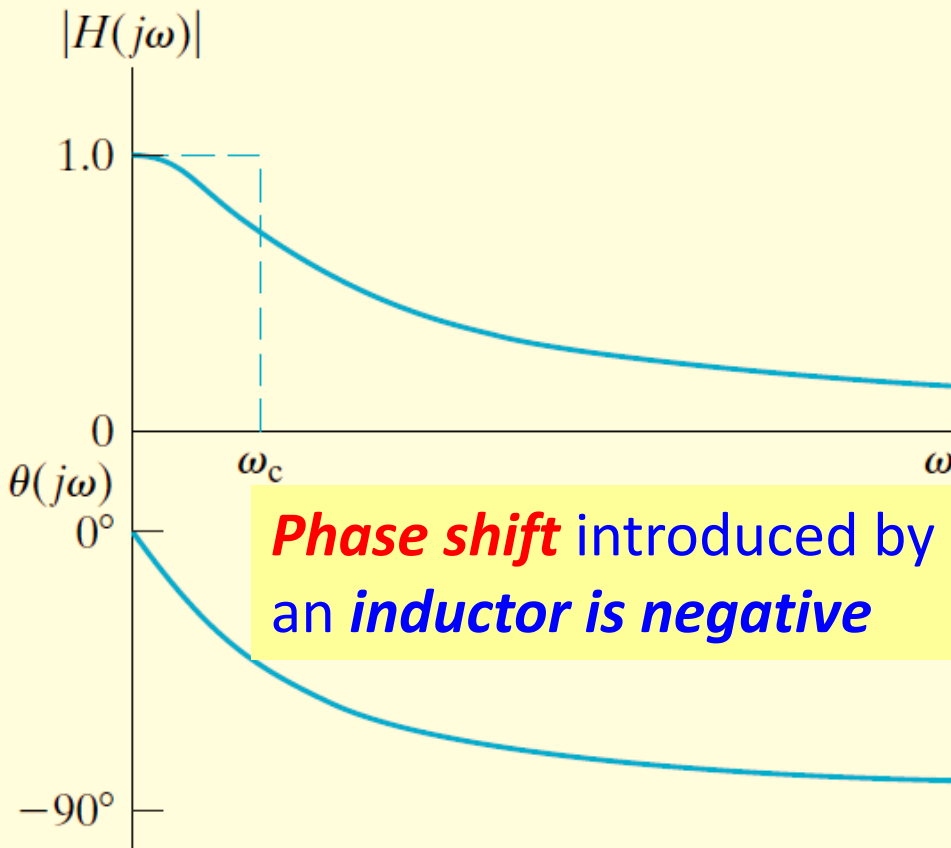
$$H(j\omega) = \frac{R/L}{j\omega + R/L}$$

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}}$$

$$\theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right)$$

To study the frequency response we make $s = j\omega$ and extract the magnitude and phase angle as function of frequency

Properties



Check the plot for $\omega=0$,
for $0 < \omega < \infty$ and
For $\omega = \infty$

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}},$$

$$\theta(j\omega) = -\tan^{-1}\left(\frac{\omega L}{R}\right).$$

Cutoff frequency of RL filters

$$|H(j\omega)| = \frac{R/L}{\sqrt{\omega^2 + (R/L)^2}},$$

$$H_{\max} = |H(j0)|,$$

From the definition
of cutoff frequency

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}}|1| = \frac{R/L}{\sqrt{\omega_c^2 + (R/L)^2}}.$$

$$\omega_c = \frac{R}{L}.$$

The Cutoff Frequency can
be set to a desired value
by selecting *R* and *L*

Meaning of Cut-off Frequency

Cutoff frequency is the frequency for which the transfer function magnitude is decreased by the factor $\left(\frac{1}{\sqrt{2}}\right)$ from its maximum value H_{max}

$$|H(j\omega_c)| = \frac{1}{\sqrt{2}} H_{\max},$$

$$P = \frac{1}{2} \frac{V_L^2}{R}.$$

$$P_{\max} = \frac{1}{2} \frac{V_{L\max}^2}{R}.$$

$$V_{L\max} = H_{\max} |V_i|.$$

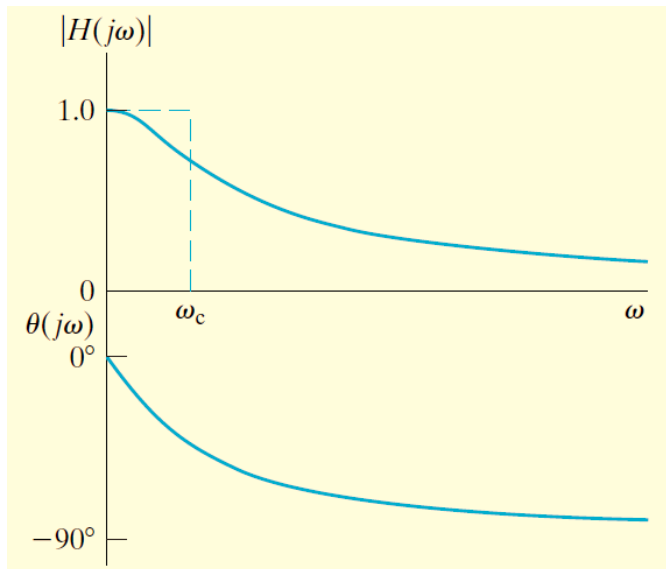
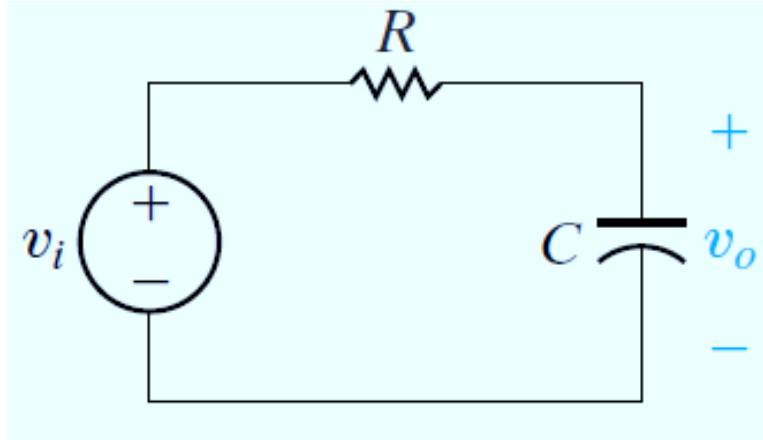
Meaning of Cut-off Frequency

$$\begin{aligned}|V_L(j\omega_c)| &= |H(j\omega_c)||V_i| \\&= \frac{1}{\sqrt{2}} H_{\max}|V_i| \\&= \frac{1}{\sqrt{2}} V_{L\max}.\end{aligned}$$

- The Cutoff Frequency is the frequency at which the *power transfer to the output is reduced to half!*

$$\begin{aligned}P(j\omega_c) &= \frac{1}{2} \frac{|V_L^2(j\omega_c)|}{R} \\&= \frac{1}{2} \frac{\left(\frac{1}{\sqrt{2}} V_{L\max}\right)^2}{R} \\&= \frac{1}{2} \frac{V_{L\max}^2/2}{R} \\&= \frac{P_{\max}}{2}.\end{aligned}$$

A Series RC Low-Pass Filter



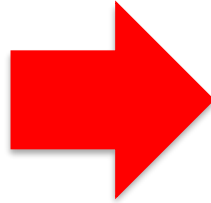
1. Zero frequency ($\omega = 0$): The impedance of the capacitor is infinite, and the capacitor acts as an open circuit. The input and output voltages are the same.

2. Frequencies increasing from zero: The impedance of the capacitor decreases relative to the impedance of the resistor, and the source voltage divides between the resistive impedance and the capacitive impedance. The output voltage is smaller than the source voltage.

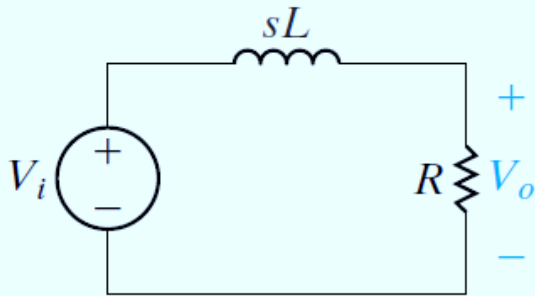
3. Infinite frequency ($\omega = \infty$): The impedance of the capacitor is zero, and the capacitor acts as a short circuit. The output voltage is zero.

Transfer Function & Cutoff frequency

$$H(s) = \frac{R/L}{s + R/L}$$

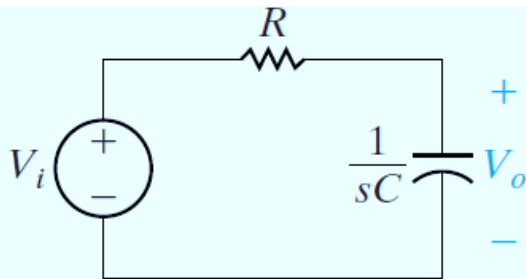
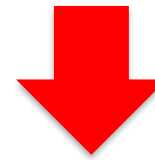


$$H(s) = \frac{\omega_c}{s + \omega_c}$$



$$H(s) = \frac{R/L}{s + R/L}$$
$$\omega_c = R/L$$

Relation of frequency domain to time domain:



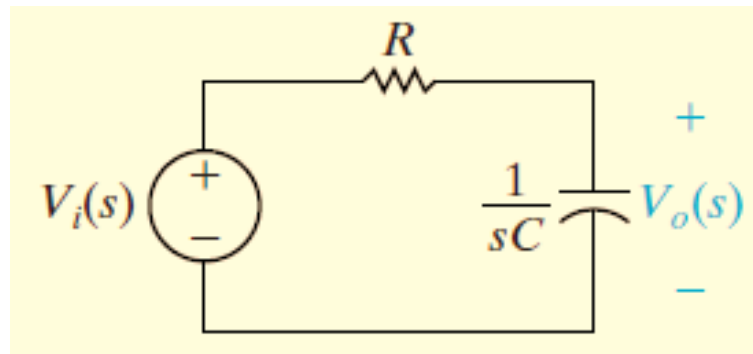
$$H(s) = \frac{1/RC}{s + 1/RC}$$
$$\omega_c = 1/RC$$

$$\tau = 1/\omega_c$$

Excercise

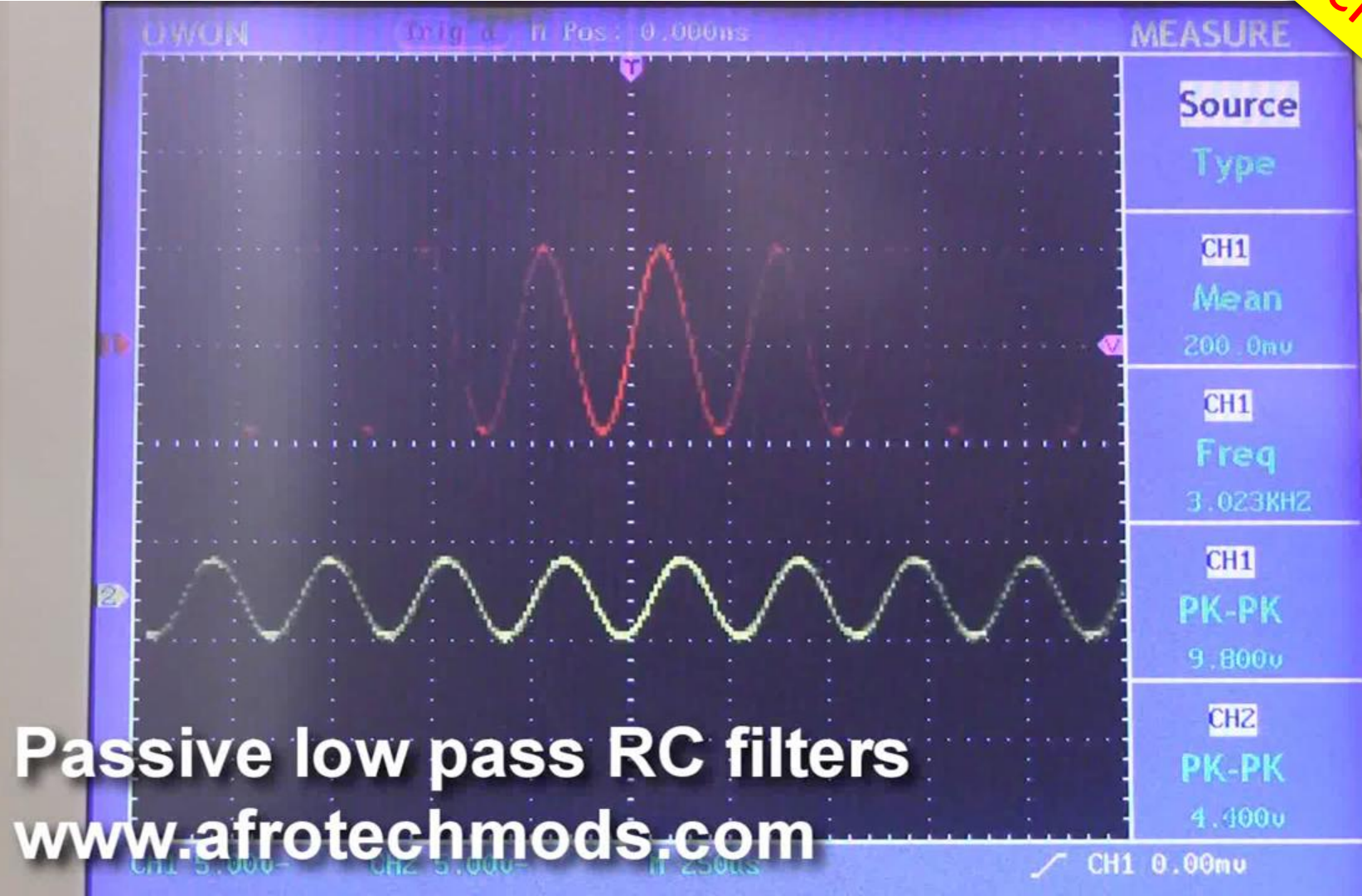
Ex. 14.2 (Nilsson)

- a) Find the transfer function between the source voltage and the output voltage.
- b) Determine an equation for the cutoff frequency in the series RC circuit.
- c) Choose values for R and C that will yield a low-pass filter with a cutoff frequency of 3 kHz.



Video Tutorial

Watch offline



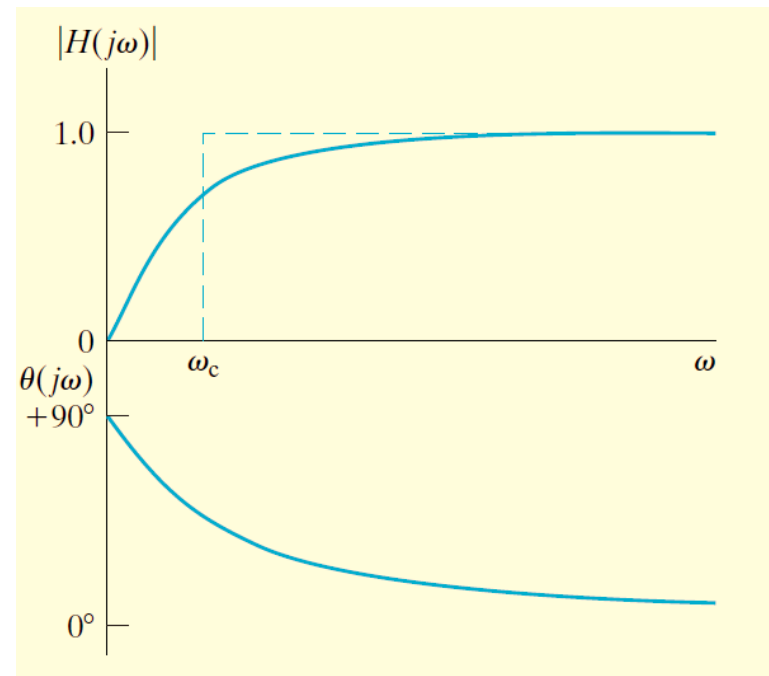
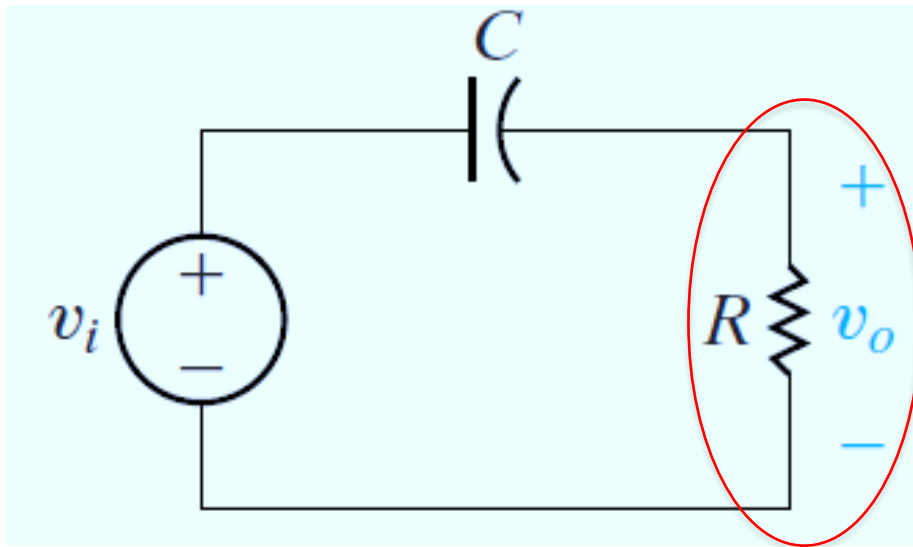
Passive low pass RC filters
www.afrotechmods.com

High-pass Filter

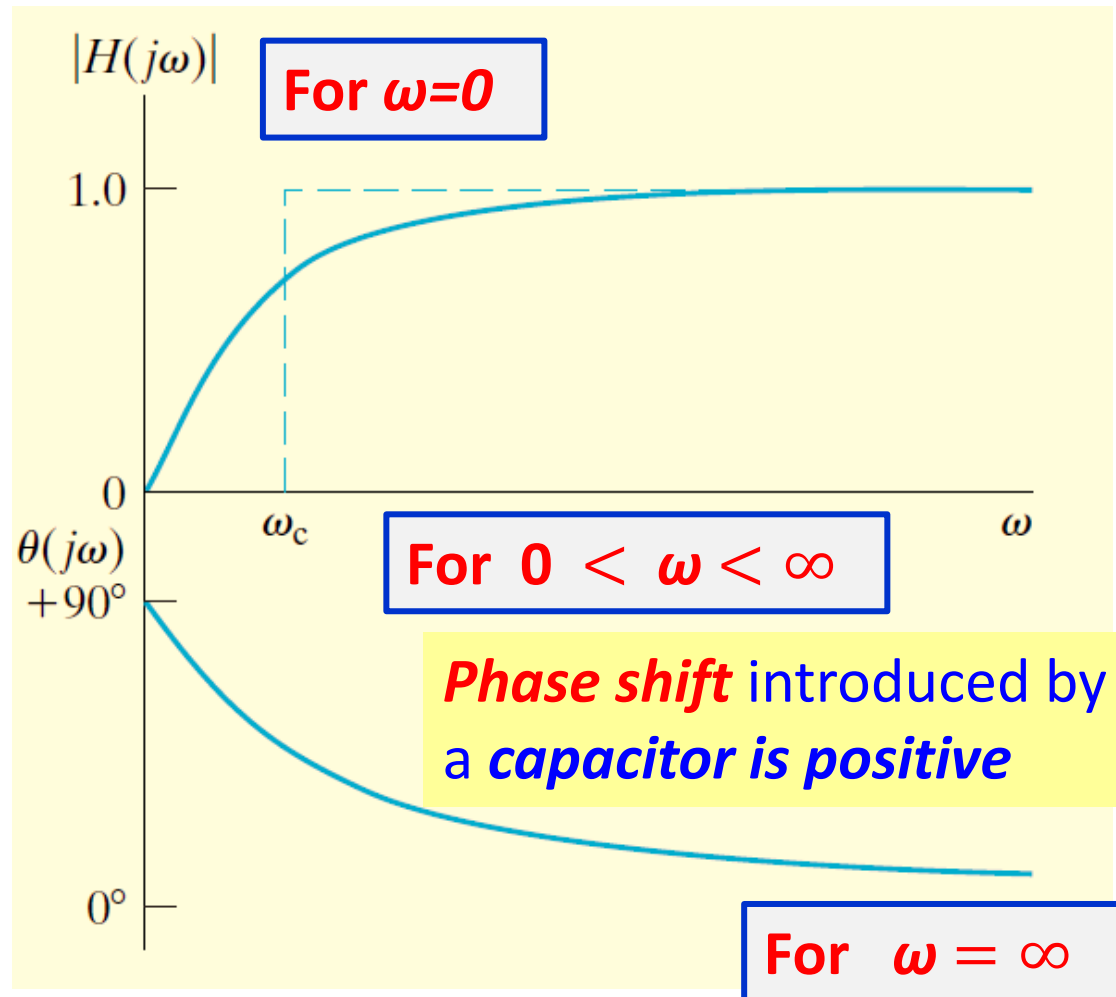
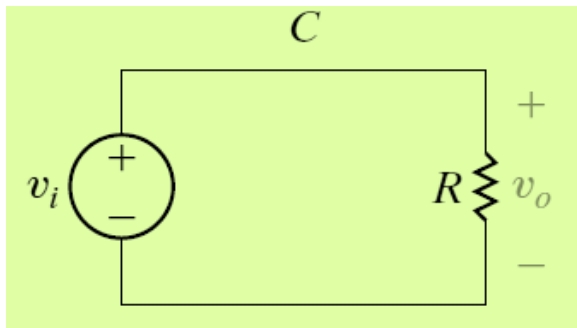
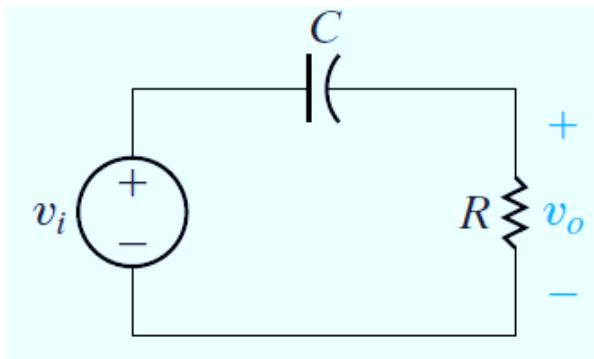
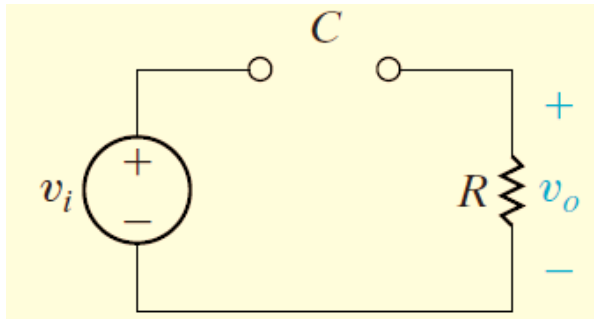
Hand notes

High-Pass Filters

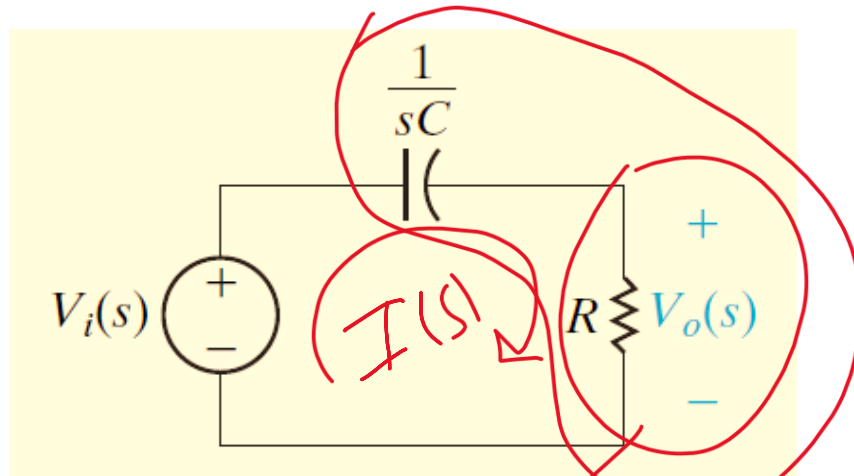
A series RC circuit which will pass high frequencies and filter out or attenuate signals of lower frequency (defined by the cutoff frequency)



High-Pass Filters



Series RC Filter: Transfer Function



$$H(s) = \frac{s}{s + 1/RC}$$

$$H(j\omega) = \frac{j\omega}{j\omega + 1/RC}$$

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + (1/RC)^2}}$$

$$\theta(j\omega) = 90^\circ - \tan^{-1}\omega RC$$

The cutoff frequency of the series RC circuit has a value of **$1/RC$** whether the circuit is configured as a Low-Pass Filter or as a High-Pass Filter

RC Filter Cutoff frequency

$$H_{\max} = |H(j\omega)|_{\omega=\infty} = |H(j\infty)|,$$

$$|H(j\infty)| = 1$$

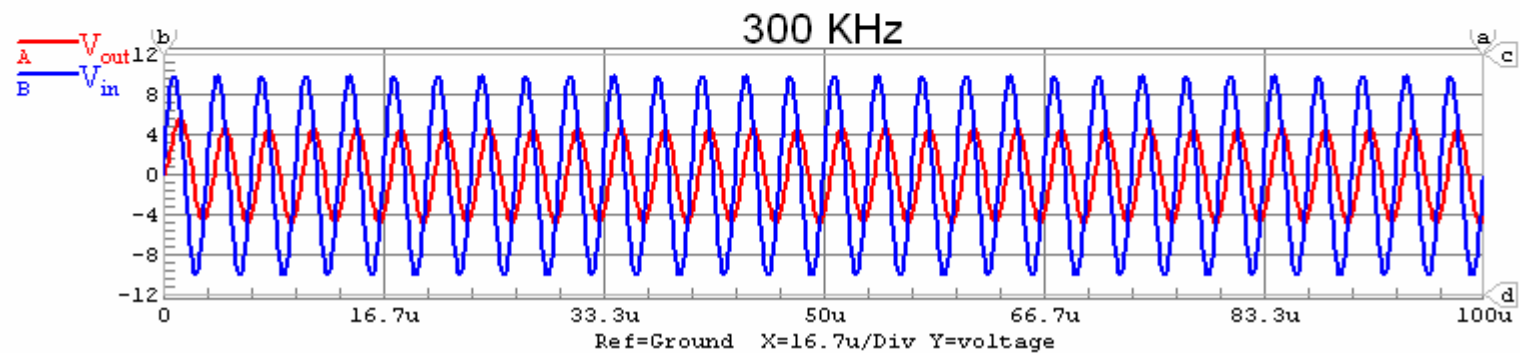
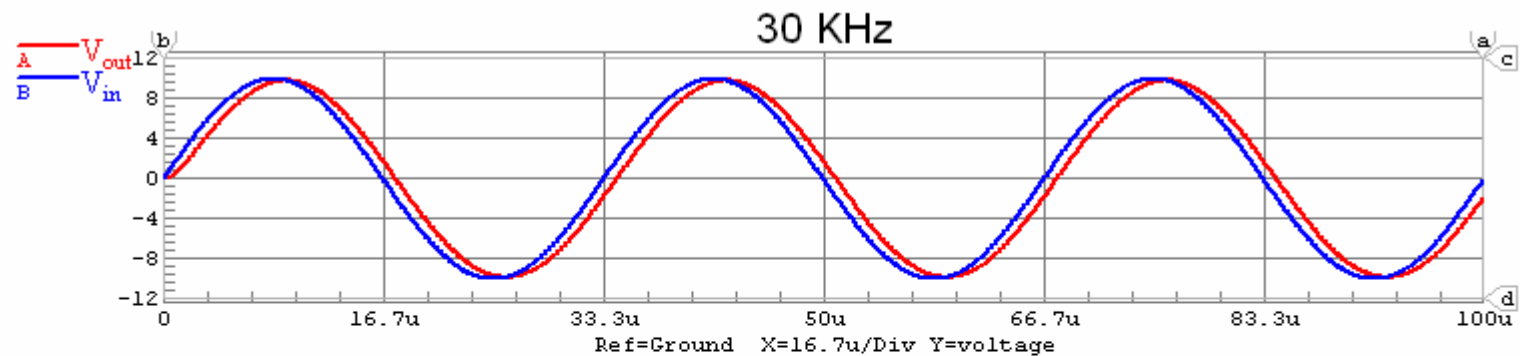
$$(1/\sqrt{2})|H(j\infty)|,$$

The filtering characteristics of a filter depends on the ***definition of the output***, circuit components and connections

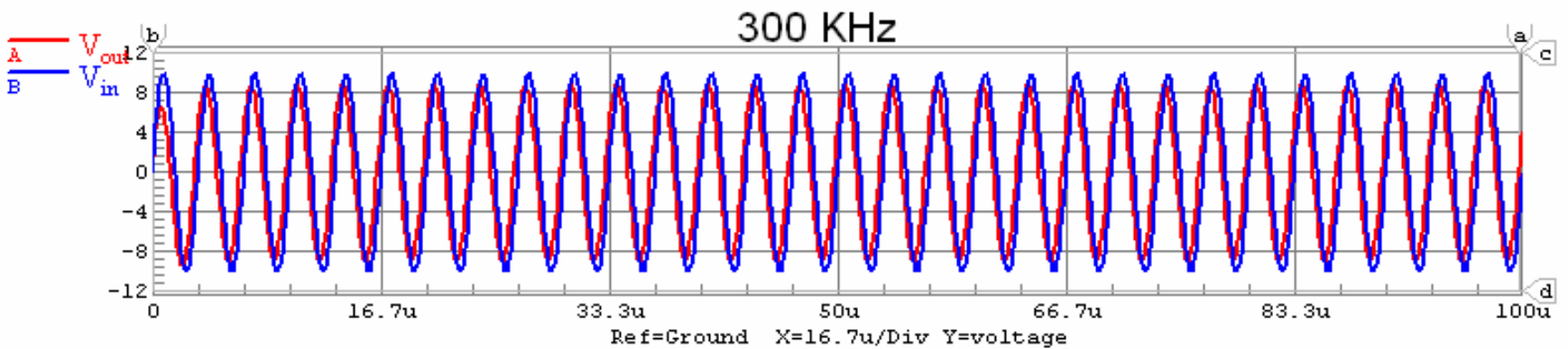
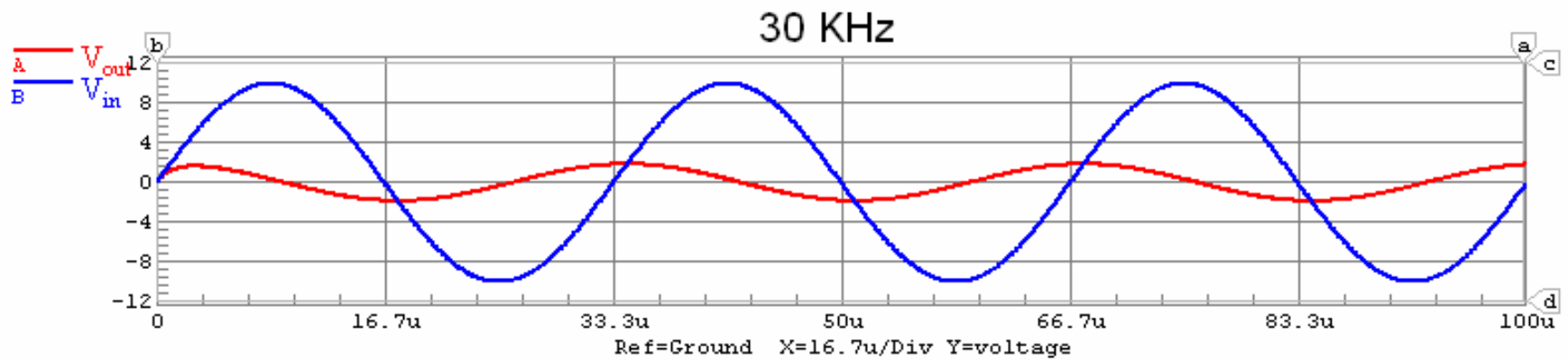
$$\frac{1}{\sqrt{2}} = \frac{\omega_c}{\sqrt{\omega_c^2 + (1/RC)^2}}.$$

$$\omega_c = \frac{1}{RC}.$$

Frequency Response

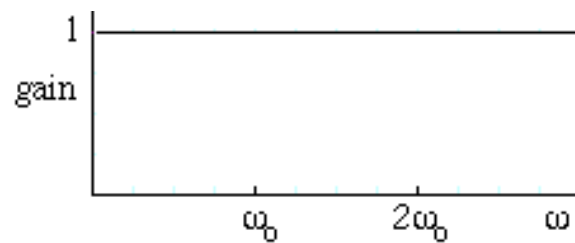
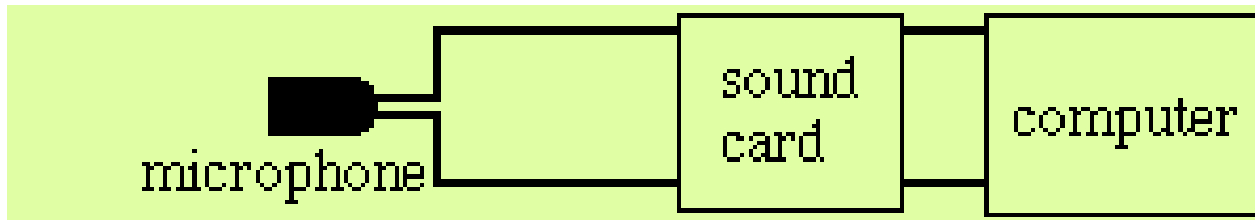


What type of filter is this?

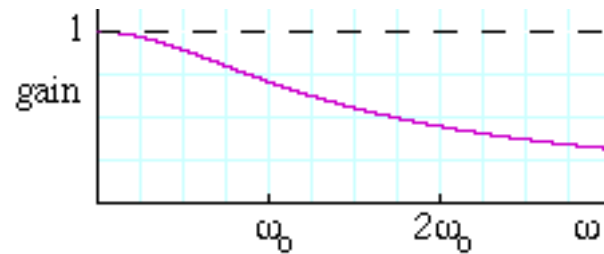


What type of filter is this?

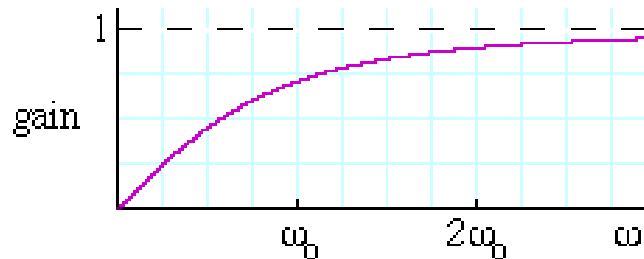
Practical Example



No filter



Lowpass filter



Highpass filter

- HP and LP this song:
«Die with a Smile»

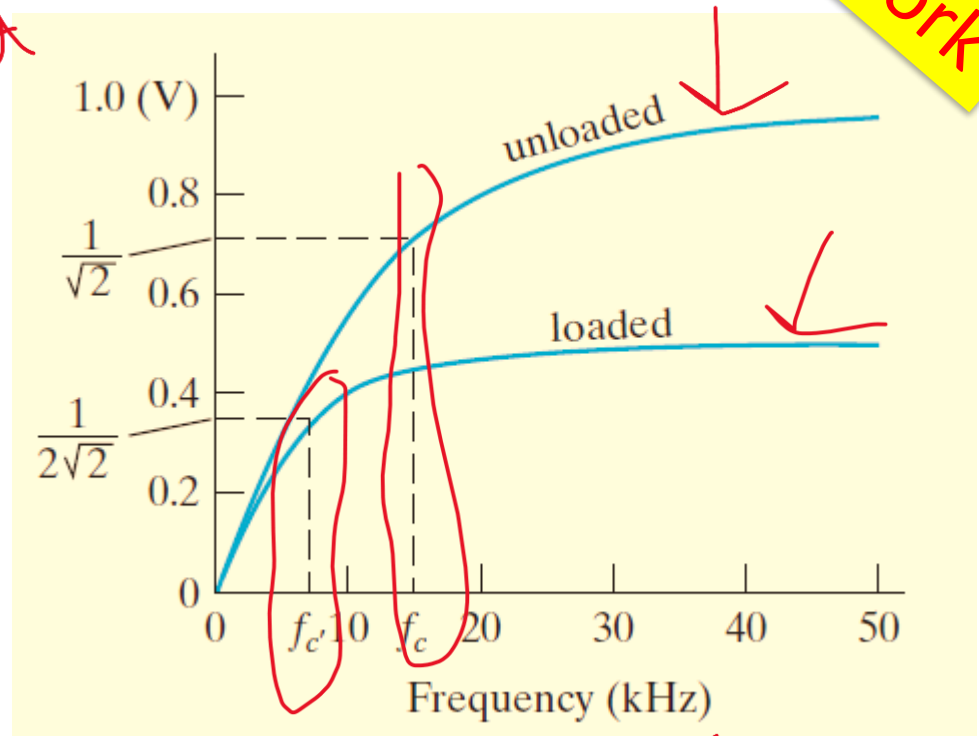
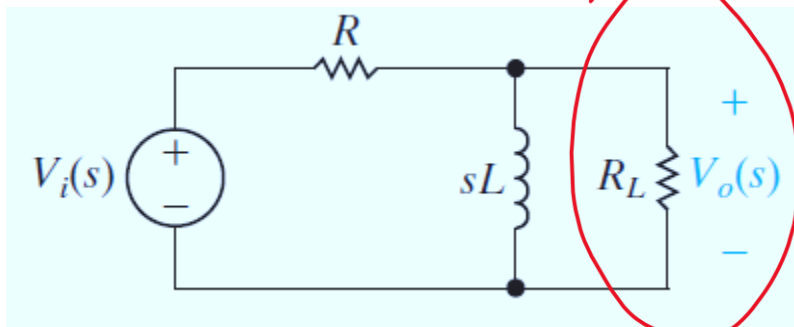
<https://editor.audio/>



Ex. 14.3 & 14.4 (Nilsson)

Homework

Add A Load



If Unloaded
 $K = 2$

$$K = \frac{R_L}{R + R_L}, \quad \omega_c = KR/L.$$

$$H(s) = \frac{\frac{R_L s L}{R_L + s L}}{R + \frac{R_L s L}{R_L + s L}} = \frac{\left(\frac{R_L}{R + R_L}\right)s}{s + \left(\frac{R_L}{R + R_L}\right)\frac{R}{L}} = \frac{Ks}{s + \omega_c},$$

Comparison

$$H(s) = \frac{Ks}{s + K(R/L)},$$

The **largest amplitude** a Passive High-Pass Filter can achieve is **1**.

Placing a load across the filter will decrease the amplitude.

$K=1$ for unloaded filter

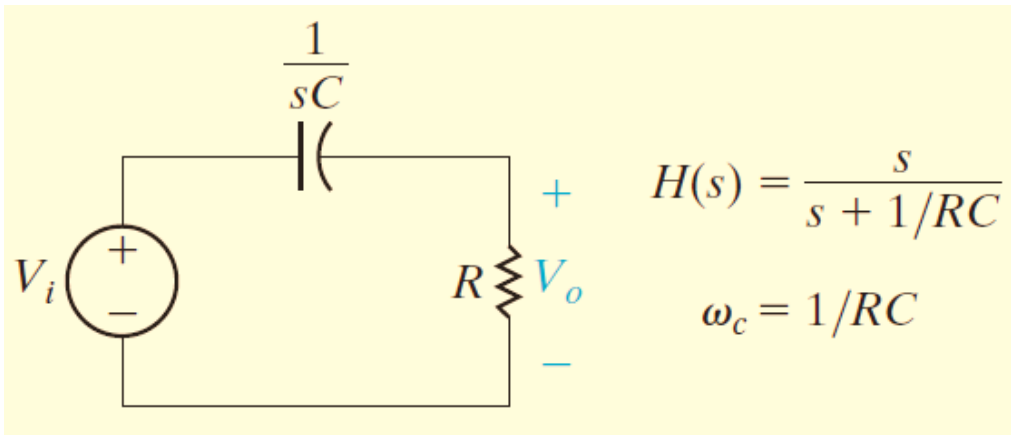
$K = R_L/(R+R_L)$ for loaded filter

$\omega_c = K(R/L)$

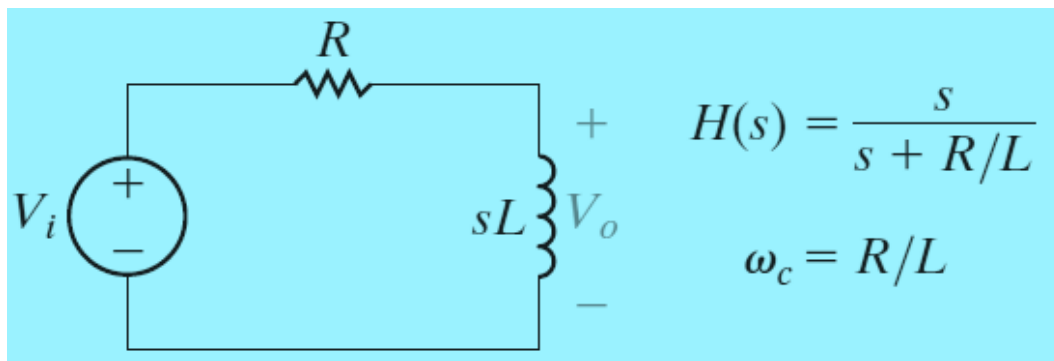
Because **$R_L/(R+R_L) < 1$** the effect of the load resistor is to reduce the passband magnitude by the factor **K** and to lower the cutoff frequency

When we need to amplify signals in the passband we must turn to Active Filters!

High-Pass Filters: Summary



$$H(s) = \frac{s}{s + \omega_c}.$$



We want a filter *transfer function to remain the same* regardless of the load. This cannot be achieved with Passive Filters

End